



VANDERBILT UNIVERSITY
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SOIL COLLECTION AND TESTING MECHANISM FOR
EXTRATERRESTRIAL MISSIONS

PROPOSAL

NASA STUDENT LAUNCH COMPETITION
2025 - 2026

SEPTEMBER 22, 2025

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1 General Information

1.1 Project Title and School Name

Project Title: *Soil Collection and Testing Mechanism for Extraterrestrial Missions*

School Name: *Vanderbilt University, Nashville, TN*

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1.5 Student Team Members

The Vanderbilt Aerospace Design Laboratory is made up of approximately 25 students in various engineering disciplines. The conclusive list of all students involved in the 2025–2026 USLI competition is the following:

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1.6 Student Team Member Roles

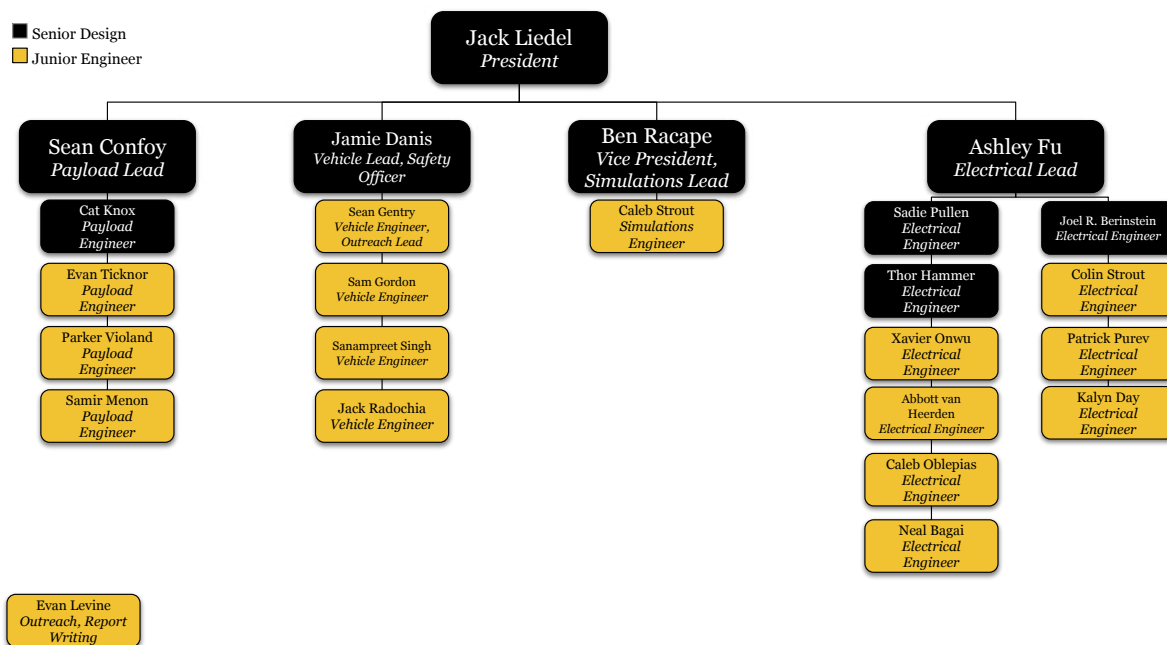


Figure 1: Student design team and organizational structure.

1.7 NAR Section Utilized During Competition

The National Association of Rocketry (NAR) section that will work in collaboration with VADL throughout this competition season is listed below. VADL will comply with all NAR High Power Safety Codes (described in Section 3.4) that are relevant to the project.

Music City Missile Club

NAR #589

<https://mc2rocketry.com/>

Organizational Structure:

- President – Peter Tarle
- Vice President – Drew Nickle
- Secretary – Allen Hall
- Treasurer – Henry Coffey
- NAR Advisor – Alex Van Bruggen

1.8 Hours Worked

To track hours worked, VADL uses a custom Excel-based system in which each team member reports their hourly contributions at the end of each day. The sheet aggregates entries across individuals and weeks to generate an accurate record of total team effort. Reported hours include both time spent actively developing subsystem hardware/software and time spent in meetings dedicated to project planning or technical discussion. As of September 21st, the team has logged 709 hours on the project.

1.9 Management Practices

NASA's USLI competition necessitates collaboration among student engineers with diverse backgrounds, skills, and experience to deliver a single integrated rocket. Given the complexity of that objective and the variance within the team, maintaining professional structures and practices is essential. Accordingly, we have refined our work organization and adopted professional tooling to foster clarity, accountability, and steady progress.

Much of our planning is derived from Agile principles. We use Jira to track progress, ownership, due dates, and key indicators. Work is organized into sprints of roughly two weeks, which may be stretched or compressed to align with major dates such as report submissions or launches. Each sprint begins with a kick-off to discuss the goals and work associated and concludes with a retrospective to capture lessons learned and refine the workflow. From a categorical standpoint, work is grouped as Jira Epics—large bodies of work like Subscale Vehicle Design, Proposal Writing, or Sensor Selection—which decompose into tasks: single-owner, verifiable items tied to a due date. Using a professional tool like Jira ensures every task has a clear owner, defined responsibilities, and a firm due date, with status and dependencies tracked end-to-end. This temporal (sprints) and categorical (epics) structure provide clear visibility into progress and bottlenecks, enabling earlier intervention and better predictability.

Our meeting rhythm works in tandem with our management structure. We hold stand-ups on Monday, Wednesday, and Friday for quick progress checks and blocker removal, and subsystem scrums approximately twice per week for deeper technical coordination. Additional reviews are held weekly with faculty and mentors to present on progress. Design decisions are requirements-driven: mission and subsystem requirements—with explicit success criteria—guide design toward lower complexity and higher reliability. We use trade studies to justify choices on a calculable basis, maintain a verification matrix to track V and V.

Lastly, mentorship and learning are deliberate. We have welcomed more junior members than in previous years and instituted mentorship pairings between underclassmen and senior team members to accelerate onboarding and knowledge transfer. The team also has run and will continue concise internal sessions on avionics, recovery, simulations, manufacturing and safety to build capability and continuity throughout the year.

2 Facilities and Equipment

2.1 Facilities and Hours of Accessibility

The VADL team has access to several facilities, each with its own unique equipment, to facilitate the design, build, and test of the vehicle and payload systems. These resources will be covered in further detail in the following sections.

Located on the fourth floor of the Olin Hall engineering building at Vanderbilt University, the Vanderbilt Aerospace Design Laboratory is the central hub for team activities. The majority of design and fabrication efforts take place in this facility, which is accessible to team members 24/7. Located in the lab are a plethora of tools, electronics, and fasteners, as well as machines and equipment used for manufacturing. The lab is also equipped with two 3D printers: a Prusa i3 MK3s (Figure 3) for producing simple PLA parts, and a Markforged Mark Two (Figure 3) for more complex, strength-critical components made with carbon fiber or onyx. These two machines provide the team with a breadth of additive manufacturing capabilities. Additionally, VADL owns a Bantam Tools desktop CNC milling machine (Figure 4). This piece of equipment enables the team to produce intricate machined parts that would otherwise be laborious to produce in the machine shop. Several team members are CNC-trained and will be the sole users of the machine. Along with the aforementioned machinery, the lab also houses a vacuum chamber, hot box, handheld power tools, electronics equipment, and a soldering station.



Figure 2: The Vanderbilt Aerospace Design Laboratory.



Figure 3: Markforged (left) and Prusa 3D printers.



Figure 4: Desktop CNC milling machine.

VADL frequently utilizes static and dynamic load frame testing in order quantify material properties. The static load frame is located within a Material Science lab on the fourth floor of Olin hall. It is capable of exerting a maximum force of 50 kN for both tensile and compressive loads. The dynamic load frame is located within the laboratory itself.



Figure 5: Static load frame.



Figure 6: Dynamic load frame.

Another facility utilized by VADL is the Engineering Conference Room in the Engineering and Science Building (ESB). Open from 9:00 AM to 5:00 PM Monday through Friday, this professional space is used for team presentations with faculty, teleconferences, and other meetings.

Vanderbilt University is home to multiple machine shops that are available to trained team members. Located in Olin Hall, Featheringill Hall, and ESB, these shops house traditional and CNC milling machines, band saws, drill presses, lathes, and plasma and laser cutters. VADL members can access the Featheringill machine shop via appointment with shop manager Phil Davis on weekdays between 11:00 AM and 5:00 PM, and graduate student mentors can provide access to the Olin Hall machine shop.

In addition to traditional machine shops, VADL also has access to specialty labs at Featheringill Hall from 9:00 AM to 5:00 PM on weekdays. The team can use the wind tunnel (Figure 7) for aerodynamic testing and flow field visualization.



Figure 7: Featheringill wind tunnel.

Located in the ESB, the Wond'ry Makerspace contains a variety of fabrication equipment. The laser cutter (Figure 8), open 1:00-9:00 PM Monday-Friday, can be used by the team for rapid production of various prototype and structural components. The Makerspace also has a Stratasys F170 printer (Figure 8) with dissolvable filaments, allowing team members to produce more complex 3D-printed part geometries. All Makerspace resources are available to VADL, provided that a Makerspace Mentor is on duty to supervise the usage of the equipment.



Figure 8: Laser cutter (left) and printer at the Wond'ry Makerspace.

2.2 Supplies

The team lab is well-stocked with most of the supplies required for vehicle and payload fabrication. VADL has decided to use fiberglass for the airframe of the rocket, which will be procured and then assembled in-house. Aluminum stock, fasteners, and all other hardware can be found in the lab as well, in addition to a wide array of electronic components that will be needed for the payload. The team has a detailed procurement process that will be enacted for any supplies not present in the lab; an in-depth inventory will be conducted prior to the fabrication cycle to ensure anything needed will be available well in advance.

2.3 Computer Equipment and Software

All digital work – including communication, vehicle and payload design, software development, and social media – will be conducted primarily on each team members' personal computers. Vanderbilt is equipped with university-wide Wi-Fi, allowing the team to work from anywhere on campus. Additional computer labs are available at libraries and other buildings across the school, and desktop computers are available in the team lab for more graphics-intensive work.

The team will utilize both Discord and Zoom for digital communication. The team Discord channel will be the primary form of communication among members, while Zoom will be used for larger meetings with alumni mentors. A shared OneDrive is used for all project documentation, providing a free and centralized repository for collaboration. Overleaf will be used for report formatting, while Jira is used for organizing tasks.

Various software tools will be employed across different aspects of the design and simulation process. The mechanical team will use SolidWorks for CAD. On the simulation side, MATLAB, OpenRocket, and RockSim will all be used in tandem for flight dynamics. JSBSim is also utilized to simulate flight dynamics for hardware in the loop testing with the payload electronics. ANSYS packages will be used for both computational fluid dynamics (CFD) and finite element analysis (FEA). Software development will be done using VS Code. A private GitHub repository will be used to store all software files, and a private network directory is shared by team members for version control management of CAD, FEA, and CFD files.

3 Safety

3.1 Safety Personnel

Every VADL member shares responsibility for safety, as affirmed in our Written Safety Statement. To maintain our strong safety culture and continuously improve it, we also appoint dedicated safety personnel, including the Student Safety Officer and subsystem safety leads, to develop, implement, and oversee policies, procedures, and checklists across design, test, and operations.

Mark Thelen, the mechanical engineer and laboratory manager, serves as the Staff Safety Officer for the VADL team. He supports safe practices at VADL by procuring necessary safety equipment, providing equipment trainings, and overseeing the reception, storage, and handling of energetic material.

Chris Dondanville serves as VADL's Range Safety Officer. He holds a NAR Level 3 certification needed to purchase high powered rocket motors. Chris has the necessary high-powered model rocketry and Student Launch Initiative experience to oversee the launch site on test days, as well as assembly and handling of VADL's solid-propellant motors and black powder ejection charges. Chris will also have the final say in any vehicle or component changes after a successful full-scale flight, as per USLI Rule 2.6.7.

Jamie Danis is a senior mechanical engineering student who will serve as the Student Safety Officer (SSO). He is an Eagle Scout with training in emergency preparedness, safe machine operation, and lifesaving and first aid. In accordance with USLI Rule 5.1, the SSO collaborates with the RSO to adapt the VADL Launch and Safety Checklist and to implement it for field flights and drop tests. Per USLI Rule 5.2, the SSO will:

- 5.2.1 Monitor team activities with an emphasis on safety during:
 - 5.2.1.1 Design of vehicle and payload.
 - 5.2.1.2 Construction of vehicle and payload.
 - 5.2.1.3 Assembly of vehicle and payload.
 - 5.2.1.4 Ground testing of vehicle and payload.
 - 5.2.1.5 Subscale launch tests.
 - 5.2.1.6 Full-scale launch tests.
 - 5.2.1.7 Competition launch.
 - 5.2.1.8 Recovery activities.
- 5.2.2 Implement procedures developed by the team for construction, assembly, launch, and recovery activities.
- 5.2.3 Manage and maintain current revisions of the team's hazard analyses, failure modes analyses, procedures, and SDS and chemical inventory data.
- 5.2.4 Assist in the writing and development of the team's hazard analyses, failure modes analyses, and procedures.

Not only is it the responsibility of the SSO to take charge of both Lab and Test safety practices, but also to consult with the applicable subject matter experts for each subcategory to corroborate best practices. The RSO is the expert who will be consulted for Test safety policy and implementation

questions. Jamie also has the privilege and opportunity to consult with Jonathan Zak, a previous VADL SSO who is currently a master's student in the lab. Mr. Zak is the Lab safety expert who will be consulted when adopting legacy VADL safety practices as well as implementing new ones, as described in the sections below.

In addition to the above duties, Jamie will hold weekly safety briefings at the weekly Friday Team meetings with all members, as well as prior to all safety-critical activities, namely vehicle and payload ground tests as well as vehicle flight tests and payload drop tests.

3.2 Hazard Analysis

In order to ensure compliance with all USLI Rules and Requirements, the SSO has researched, drafted, verified, and conducted a comprehensive safety checklist. These checklists cover two main locations work will be done: in the Lab and in the Field. The Lab encompasses VADL's dedicated lab space and other facilities where design, construction, and assembly occur. The Field refers to outdoor locations where tests and launches occur, including the launch site and ground-test site.

VADL has always placed significant emphasis on launch safety. The lab takes safety very seriously and is especially diligent in following all necessary codes, rules, and instructions from the RSO and SSO. A large portion of this success is due to well-developed and well-executed launch safety checklists, developed for both subscale and full-scale launches when the requirements for the specific launch are determined. This year, VADL has decided to implement safety checklists for lab, test, and launch procedures through a Safety Audit conducted by the SSO with consultation with VADL alumni and current team members. Benchmarks for Lab safety and Ground testing were developed by the SSO in his effort to carry out his outlined duties in USLI Rule 5.2 (to ensure fabrication, testing, and general lab safety) through preparedness. This preparedness includes procuring and maintaining necessary safety equipment as well as implementing and enforcing safe processes and norms. The hazards identified and mitigation strategies implemented are documented in the example Hazard Analysis Matrix in the section below. As the team approaches subscale and fullscale launch, the Launch Safety Checklist will be further developed in order to further detail the mitigation strategies following the Example Hazard Analysis Matrix.

VADL plans to use the table below to categorize and identify the risk associated with any hazard that comes up in the design, construction, assembly, testing, or launch of our vehicle and payload. In this assessment, any given hazard's severity is weighed equal to its likelihood of occurring during the entire lifecycle of the project. In the table below, hazards are documented with their identified Risk Score before the Lab Safety Audit, and at the time of Proposal submission, with future critical safety work noted as to be completed and documented prior to PDR submission. The team intends to use the below Human Hazard Analysis both to track the progress throughout the entire project lifecycle as well as a basis Hazard Analysis template from which to extrapolate more in-depth launch, personal, failure, and environmental Hazard Analyses in future reports. For reference, a Catastrophic event involves death, serious injury, or serious property damage such as an uncontrolled building fire. A Negligible event involves small injury or property damage that can be healed or fixed within a day, such as a paper cut.

Table 1: Preliminary Personnel Hazard Analysis Matrix

Hazard for People / Setting	Causes	Risk Level At Safety Audit (8/20/25)	Preparedness Strategy	Mitigation Strategy	Status at Proposal	Risk Level At Proposal (9/21/25)	Future Responsibility
Minor Lacerations/Burns (In-Lab)	1. Hand tool misuse 2. Soldering iron routine use	2B	* Two wall-mounted lab first aid kits	1. Common hand tool training during initial Safety Briefing 2. Soldering iron training and cleanup held by electrical team lead	Fully Prepared Fully Mitigated	3C	* Check and restock first aid kit bimonthly (Jamie Danis)
Major Lacerations (In-Lab)	1. Power tool misuse (Reciprocating Saw, Miter Saw, Angle Grinder, Mill, Lathe) 2. Disorganized/hazardous scrap metal bin	1D	* Severe-injury first aid kit mounted and visible * 7 min walk / 2 min drive to VU Emergency Dept.	1. Instituted and enforced Tool Qualification Punchcards 2. Completely cleaned, reorganized, and de-hazardized scrap metal racking	Fully Prepared Partially Mitigated	2E	* Continue power tool qualification trainings and enforcement (Jamie Danis)
Uncontrolled Fire (In-Lab)	1. Haphazard power cable management 2. Improper LiPo Battery storage 3. Improper energetic material storage	1D	* 2 wall-mounted fire extinguishers in-date and visible	1. Organize and execute lab cleanup 2. Keep LiPo batteries in fire-resistant, pressure-relieved, damage-resistant steel containers 3. Energetic materials stored in explosion-rated containers, and only handled and stored by Staff Safety Officer, Mark Thelen	Fully Prepared Fully Mitigated	2E	* Continue energetic material personnel restriction (Mark Thelen) * Enforce continuous use of new LiPo battery storage procedure (Jamie Danis)
Trips/Falls (In-Lab)	1. Disorganized scrap material area 2. Overcluttered lab floor	3B	* First aid kits * Vanderbilt Health Services	1. Complete cleanup of scrap material area 2. Lab-wide cleanup and relocation of clutter to dumpster or storage	Fully Prepared Fully Mitigated	3D	* Hold future cleanups as needed (Jamie Danis)
Hazardous Dust Inhalation (In-Lab)	1. Fabrication (Composites, wood) 2. In-Lab composite grinding	2B	* Respirators and replacement filters acquired and located throughout lab in visible and accessible locations * Sawhorses built and available for outdoor composite grinding	1. Portable HEPA filtration unit tested and located at Indoor Composite Fabrication Workstation 2. All composite grinding and sanding done strictly outdoors (policy enforced, and outdoor alternative made available)	Fully Prepared Partially Mitigated	2C	* Build and enforce outdoor composite workstation (Jamie Danis)
Skin/Eye Irritation (In-Lab)	1. Haphazard composite sheet storage 2. Insufficient PPE availability/visibility 3. Fabrication (Composites)	3B	* Topical analgesic available in lab first aid kits * Handwash and eyewash sink available in lab	1. Dedicated, closed containers for composite materials 2. PPE acquired and distributed visibly 3. Dedicated composite fabrication workstation with posted PPE and mitigation policies	Fully Prepared Partially Mitigated	4C	* Designate outdoor composite workstation and post procedures (Jamie Danis)

Electrical Shock (In-Lab)	1. Misuse of corded power tools 2. Insufficient experience when preparing payload electrical system	2D	* First aid kits * Vanderbilt Health Services	1. Use of cordless power tools 2. Electrical work over 12V restricted to dedicated, trained electrical team	Fully Prepared Fully Mitigated	3E	* Continue to enforce electrical safety (Ashley Fu, Jamie Danis)
Soldering Fume Inhalation (In-Lab)	1. Unhelpful Helping Hands setup 2. Lack of proper fume-filtering fan setup	3B	* Organic Vapor respirators available and visible	1. High-quality helping-hands acquired 2. Fume fan acquired	Fully Prepared Fully Mitigated	4C	* Continue to enforce soldering safety (Ashley Fu, Jamie Danis)
Energetic Material Compustion (In-Lab)	1. Improper/Irresponsible energetic material storage 2. Accident while preparing ejection charges	2C	* Dedicated energetic material location in storage room, and in lab when actively working on them	1. Strictly enforced policy of motors only handles by Staff Safety Officer 2. Ejection charges only prepared by trained personnel	Fully Prepared Fully Mitigated	2E	* Continue to enforce energetic material restrictions (Mark Thelen, Jamie Danis)
Exhaustion/Fatigue/Mental Anguish (In-Lab)	1. Demanding work hours 2. Pressure to perform while tired	3A	* Policies regarding mental wellbeing prioritization posted and enforced	1. Zero tolerance policy for operating tools while significantly tired or altered in any way 2. Everyone in the team commits to respect the mental wellbeing prioritization of others	Fully Prepared Partially Mitigated	3C	* Continue to implement mental health prioritization throughout the year, especially during high-stress periods (All Team Members)
Minor Lacerations/Burns (Ground Test)	1. Insufficient Safe Distance observed during ground test	3D	* Ground Test Safety Kit assembled, labeled and posted at lab exit	1. Ground Test Safety Kit used to cordon off a Safe Zone for drop and ejection tests	Fully Prepared Partially Mitigated	3E	* Implement Ground Test Safety Kit on first ground test and troubleshoot if necessary (Jamie Danis)
Fire (Ground Test)	1. Ground test conducted in dry/grassy area	2D	* Ground Test Safety Kit assembled, labeled and posted at lab exit	1. Ground Test Safety Kit includes fire extinguisher and water for site-dousing	Fully Prepared Partially Mitigated	3E	* Implement Ground Test Safety Kit on first ground test and troubleshoot if necessary (Jamie Danis)

Minor Laceration/Burn (Launch)	1. Mistake when preparing vehicle/payload 2. Fall risks inherent to outdoor work	3C	* Field first-aid kit inspected and prepared	1. All field tasks strictly overseen by RSO, acknowledged with complete authority over launch 2. Proper shoes and outerwear will be worn	Fully Prepared Partially Mitigated	4D	* Obey all safety procedures and instructions upon first launch, troubleshoot if necessary (All Team Members, Jamie Danis)
Major Laceration (Launch)	1. Improper Safe distance observed during launch 2. Improper launch site clearance observed	1D	* Field first-aid kit inspected and prepared * Multiple present team members trained in life-saving measures	1. NAR High Power Rocket Safety Code Minimum Safe Distance strictly followed 2. All field tasks strictly overseen by RSO, acknowledged with complete authority over launch	Fully Prepared Partially Mitigated	2E	* Obey all safety procedures and instructions upon first launch, troubleshoot if necessary (All Team Members, Jamie Danis)
Fire/Major Burn (Launch)	1. Improper launchsite choice and ground preparation 2. Motor CATO	1D	* Launch Go-Bag includes fire extinguisher and water for site-dousing	1. Launchsite chosen to comply with NAR High Power Rocket Safety Code 2. All energetic material transported in explosive-rated steel containers exclusively by Mark Thelen	Fully Prepared Partially Mitigated	2E	* Obey all safety procedures and instructions upon first launch, troubleshoot if necessary (All Team Members, Jamie Danis)

3.3 Chemical and Energetic Safety - Safety Data Sheets

VADL has a long history of safe storage, handling, and use of composites, adhesives, energetics, and other hazardous materials and compounds. The safety data sheets for these materials are available to all team members in a labeled physical binder in their cabinet as well as online. Above this cabinet, gloves and respirators are kept in stock and plainly visible, with signage requiring their use. Below are attached the materials and compounds currently in stock as well as those anticipated to be used in this project. When the payload and vehicle designs are more developed in later NASA reports, the Safety Data Sheet compilation will be updated with all new potentially hazardous materials.

Energetics:

- Aerotech I366 Motor: <https://tinyurl.com/pvhme673>
- Cesaroni L2375 Motor: <https://tinyurl.com/mspeeum9>
- Owen Compliance Black Powder: <https://tinyurl.com/3peubrha>

Composite Sheets:

- Fibre Glast Part #262 4 oz. Fiberglass Fabric: <https://tinyurl.com/8st7s7km>
- Fibre Glast Part #1069 - 5.7oz Graphite 3K 2 x 2 Twill Carbon Fiber: <https://tinyurl.com/dkfyepdd>

Epoxy:

- Devcon 2 Ton Epoxy [1:1]: 2-Ton Hardener, Extended worktime <https://tinyurl.com/msdhahr5>
- Devcon 5 Minute Epoxy amber [1:1]: 5-Minute Epoxy Resin / Hardener <https://tinyurl.com/9w798jje>
- Fibre Glast Part #2000 System 2000 Epoxy Resin: <https://tinyurl.com/uawndhzn>
- Fibre Glast Part #2120 Epoxy Hardener: <https://tinyurl.com/377hpja9>
- J-B Weld Steel Reinforced Epoxy Resin and Hardener: <https://tinyurl.com/54djueuj>

Glues/Adhesives:

- Bob Smith Industries – Maxi-Cure Thick Cyanoacrylate Glue: <https://tinyurl.com/26j4pyay>
- Bob Smith Industries – Cyanoacrylate Accelerator: <https://tinyurl.com/vjayej5e>
- 3M Super 77 Adhesive, Bulk: <https://tinyurl.com/3wdsvpks>
- Loctite 242 Threadlocker: <https://tinyurl.com/ms3ck2tr>
- 3M Polyurethane Foam Insulation Adhesive 78 ET <https://tinyurl.com/yc7tb46c>

Solvents:

- Ethyl Alcohol, 100%, Decon Labs: <https://tinyurl.com/mu6fzhaw>
- Klean-Strip Acetone: <https://tinyurl.com/ys6bk5j3>
- Sigma Aldrich Hydrogen Peroxide: <https://tinyurl.com/24mx29v>

Mold Release Compounds:

- Fibre Glast Part #1153 - FibRelease: <https://tinyurl.com/yv3pnnnz>
- Fibre Glast Part #13 – PVA Release Film: <https://tinyurl.com/4jp4zbeb>
- Meguiar's M08, Maximum Mold Release Wax (23-135A): M0811: <https://tinyurl.com/myfk44du>

3.4 Laws and Regulations

In order to ensure rocket flight safety for all VADL members as well as our launch vehicle, applicable National Association of Rocketry, Federal Aviation Commission, and Tennessee state laws have been compiled below. These laws and regulations are referenced in the Written Safety Statement for all VADL team members to review prior to the submission of this Proposal.

National Association of Rocketry (NAR)

The NAR High Power Rocket Safety Code (HPRSC) is to be reviewed in its entirety and followed by all VADL team members.

A physical copy is printed and available in the VADL office.

A digital copy is stored in the VADL shared folder.

A copy of the Launch Minimum Distance Table (included below) will be posted and enforced at each launch.

Though all 13 Sections of the NAR HPRSC are crucial to read and understand in their entirety, those most pertinent to our anticipated launch vehicle and most pressing to the safety of our members are copied below.

Certification: I will only fly high power rockets or possess high power rocket motors that are within the scope of my user certification and required licensing.

Launch Safety: I will use a 5-second countdown before launch. I will ensure that a means is available to warn participants and spectators in the event of a problem. I will ensure that no person is closer to the launch pad than allowed by the accompanying Minimum Distance Table. When arming onboard energetics and firing circuits I will ensure that no person is at the pad except safety personnel and those required for arming and disarming operations. I will check the stability of my rocket before flight and will not fly it if it cannot be determined to be stable. When conducting a simultaneous launch of more than one high power rocket I will observe the additional requirements of NFPA 1127.

Source: NAR High Power Rocketry Safety Code

Energetic Materials

Energetic materials, including but not limited to the commercially available Ammonium Perchlorate Composite Propellant (APCP) motors VADL uses exclusively (as per USLI Requirement 2.7), are strictly regulated.

The safe and legal acquisition of these motors is governed by Code of Federal Regulations (CFR) Title 27, Part 555: Commerce in Explosives.

In this CFR code, APCP motors (classified as propellant-actuated devices) with more than 62.5 g of propellant are not exempt from federal regulations.

The safe handling, transportation, and installation of these motors are governed by the National Fire Protection Association (NFPA) 1127 Code for High Power Rocket Motors. A copy of this code is available in the VADL Lab office.

Compliance controls

1. Rocket motors are purchased only by experienced, trained, Level 3 certified team mentors.
2. Motors are stored in MAGloc explosives-storage magazines by the Staff Safety Officer (Mark Thelen).
3. Energetic materials are removed from storage only as needed and only by the Staff Safety Officer prior to launch.

Supervision and process controls

1. The Staff Safety Officer, Student Safety Officer, and VADL alumni mentors supervise the assembly and loading of all motors, blast charges, and electric matches.
2. Strict procedures, appropriate PPE, checklists, and verification steps are followed at every stage.

Federal Aviation Administration (FAA)

A copy of all relevant Federal, state, and local laws will be printed and available in the Lab Safety Binder. VADL will strictly follow all relevant regulations regarding unmanned rocket launches and motor handling.

Any VADL launch will be scrubbed if the following launch site conditions are met, as per the FAA 14 CFR, Subchapter F, Part 101, Subpart C (Amateur Rockets):

High-power rocket flights may not be conducted at altitudes with more than five-tenths cloud coverage or less than five miles of visibility, into any cloud, between sunset and sunrise without FAA authorization, within 9.26 km (5 nm) of an airport boundary without FAA authorization, or in controlled airspace without FAA authorization. Operations must maintain the greater of one-quarter the maximum altitude or 457 m (1,500 ft.) from any person or property not associated with the launch. A responsible adult (18+) with final flight approval must be present, and reasonable fire precautions must be in place.

Source: FAA 14 CFR, Subchapter F, Part 101, Subpart C, 101.25

Tennessee and Local Laws

Tennessee fire marshals have adopted the NFPA Code as law, which is already followed by VADL.

Collection of materials from a state park or Tennessee natural area requires a prior permit.

VADL is not launching in these areas, so no permit is required.

Source: TN Fire Marshal Adopted Codes

As proof of diligence, the team will avoid collection from restricted areas and clean soil off shovels after launch, both for material integrity and to reduce risks of transporting organic material.

Team Compliance

All team members will be fully briefed on the above laws, codes, and standards at the first Team Safety Briefing.

Team members will be reminded of the applicable codes prior to each launch.

It is the responsibility of Jamie Danis (Student Safety Officer) to facilitate the education of all VADL members on these codes, and enforce compliance both in the lab and in the field.



NAR High Power Rocket Safety Code

Approved March 2006

1. **Certification.** I will only fly high power rockets or possess high power rocket motors that are within the scope of my user certification and required licensing.
2. **Materials.** I will use only lightweight materials such as paper, wood, rubber, plastic, fiberglass, or when necessary ductile metal, for the construction of my rocket.
3. **Motors.** I will use only certified, commercially made rocket motors, and will not tamper with these motors or use them for any purposes except those recommended by the manufacturer. I will keep smoking, open flames, and heat sources at least 25 feet away from these motors.
4. **Ignition System.** I will launch my rockets with an electrical launch system, and with electrical motor igniters that are installed in the motor only after my rocket is at the launching or prepping area. My launch system will have a safety interlock that is in series with the launch switch that is not installed until my rocket is ready for launch, and will use a launch switch that returns to the "off" position when released. If my rocket has onboard ignition systems for motors or recovery devices, these will have safety interlocks that interrupt the current path until the rocket is at the launch pad.
5. **Misfires.** If my rocket does not launch when I press the button of my electrical launch system, I will remove the launcher's safety interlock or disconnect its battery, and will wait 60 seconds after the last launch attempt before allowing anyone to approach the rocket.
6. **Launch Safety.** I will use a 5-second countdown before launch, and will ensure that everyone in the launch site is paying attention and that no person is closer to the launch pad than allowed by the accompanying Minimum Distance Table. I will check the stability of my rocket before flight and will not fly it if it cannot be determined to be stable.
7. **Launcher.** I will launch my rocket from a stable device that provides rigid guidance until the rocket has attained a speed that ensures a stable flight, and that is pointed to within 20 degrees of the vertical. If the wind speed exceeds 5 miles per hour I will use a launcher length that permits the rocket to attain a safe velocity before separation from the launcher. I will use a blast deflector to prevent the motor's exhaust from hitting the ground. I will ensure that there is no dry grass within a clear distance of each launch pad determined by the accompanying Minimum Distance table, and will increase this distance by a factor of 1.5 if the rocket motor being launched uses titanium sponge in the propellant.
8. **Size.** My rocket will not contain any combination of motors that total more than 40,960 N-sec (9208 pound-seconds) of total impulse. My rocket will not weigh more at liftoff than one-third of the certified average thrust of the high power rocket motor(s) intended to be ignited at launch.

9. **Flight Safety.** I will not launch my rocket at targets, into clouds, near airplanes, or on trajectories that take it directly over the heads of spectators or beyond the boundaries of the launch site, and will not put any flammable or explosive payload in my rocket. I will not launch my rockets if wind speeds exceed 20 miles per hour. I will comply with Federal Aviation Administration airspace regulations when flying, and will ensure that my rocket will not exceed any applicable altitude limit in effect at that launch site.
10. **Launch Site.** I will launch my rocket outdoors, in an open area where trees, power lines, buildings, and persons not involved in the launch do not present a hazard, and that is at least as large on its smallest dimension as one-half of the maximum altitude to which rockets are allowed to be flown at that site or 1500 feet, whichever is greater.
11. **Launcher Location.** My launcher will be at least one half the minimum launch site dimension, or 1500 feet (whichever is greater) from any inhabited building, or from any public highway on which traffic flow exceeds 10 vehicles per hour, not including traffic flow related to the launch. It will also be no closer than the appropriate Minimum Personnel Distance from the accompanying table from any boundary of the launch site.
12. **Recovery System.** I will use a recovery system such as a parachute in my rocket so that all parts of my rocket return safely and undamaged and can be flown again, and I will use only flame-resistant or fireproof recovery system wadding in my rocket.
13. **Recovery Safety.** I will not attempt to recover my rocket from power lines, tall trees, or other dangerous places, fly it under conditions where it may recover in spectator areas or outside the launch site, or attempt to catch it as it approaches the ground.

MINIMUM DISTANCE TABLE				
Installed Total Impulse (N-sec)	Equivalent Motor Type	Minimum Clear Distance (ft.)	Minimum Personnel Distance (ft.)	Minimum Personnel Distance (Complex Rocket) (ft.)
0 - 320.00	H or smaller	50	100	200
320.01 - 640.00	I	50	100	200
640.01 - 1280.00	J	50	100	200
1280.01 - 2560.00	K	75	200	300
2560.01 - 5120.00	L	100	300	500
5120.01 - 10,240.00	M	125	500	1000
10,240.01 - 20,480.00	N	125	1000	1500
20,480.01 - 40,960.00	O	125	1500	2000

Note: A complex rocket is one that is multi-staged or that is propelled by two or more rocket motors

Figure 9: National Association of Rocketry - High Power Rocket Safety Code

3.5 Safety Briefings and Written Agreement

The preliminary VADL safety briefing took place 9/15/25. The team was briefed on all new and existing safety procedures and infrastructure described in this report, and each team member read and signed the Written Safety Statement attached below. The team was also introduced to the VADL Tool Qualification documents and process and was read the NAR High Power Rocket Safety Code.

Future safety briefings (weekly on Friday team meetings) will include coverage of new hazardous materials in the lab, address safety concerns noticed by the SSO, or give notice about upcoming launches or tests and the necessary prior precautions to be taken by each team member.

Safety briefings will also be held before each ground test and launch (subscale or full-scale). These briefings will be planned and executed by the SSO and RSO, and will cover the NAR safety code, emergency preparedness procedures, and the authority of the RSO to postpone or cancel a launch for safety reasons.

Vanderbilt Aerospace Design Laboratory - Written Safety Agreement

As a member of the Vanderbilt Aerospace Design Laboratory, I agree to abide by all posted (physically and digitally) safety procedures as well as all verbal safety-related instructions given in lab and in the field by the Student Safety Officer (Jamie Danis), Range Safety Officer (Chris Dondanville), or Staff Safety Officer (Mark Thelen). This includes, but is not limited to, PPE usage, power tool qualification, and personal relocation.

Before each rocket launch, range safety inspections will be conducted by the Range Safety Officer. The Range Safety Officer holds authority to inspect the design and construction of the vehicle and payload, and has the final authority to scrub launches based on any concerns related to the safety of all personnel and the launch vehicle. Failure to comply with the determination of the Range Safety Officer may lead to the disqualification of our team from the NASA University Student Launch program.

I also agree to familiarize myself with, and to abide by:

1. All NASA USLI safety rules [2025-2026 NASA Student Launch Handbook, Rules, Sect. 5]
2. All NAR (National Association of Rocketry) safety standards and requirements [NAR High Power Rocket Safety Code]
3. All applicable FAA (Federal Aviation Administration) regulations and policies [FAA 14 CFR, Subchapter F, Part 101, Subpart C]
4. All pertinent Tennessee State Laws regarding high-powered model rocketry flight [NFPA 1127 Code for High Power Rocket Motors]
5. All guidelines and warnings set out in the compiled Safety Data Sheets for all hazardous material in lab (found in the shared Safety folder and under the fume hood in the laboratory).

My signature below indicates that I fully understand, acknowledge, and accept all of the above. I acknowledge that failure to abide by the above may lead to my exclusion from certain tasks in the laboratory, a revocation of my permission to attend launches, or the revocation of VADL's permission to launch our rocket in the NASA USLI Competition. I acknowledge that my failure to sign and agree to abide by the above rules will result in my removal from the VADL program.

Signed:

- Neal Bagai 
- Joel Berinstein 

- Jamie Danis Jamie Danis
- Kalyn Day Kalyn Day
- Sean Confoy Sean Confoy
- Ashley Fu Ashley Fu
- Sean Gentry Sean Gentry
- Samantha Gordon Samantha Gordon
- Thor Hammer Thor Hammer
- Catherine Knox Catherine Knox
- Jack Ladd Jack Ladd
- Evan Levine Evan Levine
- Samir Menon Samir Menon
- Caleb Oblepias Caleb Oblepias
- Xavier Onwu Xavier Onwu
- Patrick Purev Patrick Purev
- Sadie Pullen Sadie Pullen
- Jack Radochia Jack Radochia
- Ben Racapé Ben Racapé
- Sanampreet Singh Sanampreet Singh
- Colin Strout Colin Strout
- Caleb Strout Caleb Strout
- Evan Ticknor Evan Ticknor
- Abbott Van Heerden Abbott Van Heerden
- Parker Violand Parker Violand

Figure 10: Written Safety Agreement signed by each VADL team member

4 Technical Design

4.1 Introduction

4.1.1 Payload Requirements (Internal and External)

The payload designed by the Vanderbilt Aerospace and Design Laboratory (VADL) is required to autonomously collect and analyze a soil sample of at least 50 mL within 15 minutes of landing. Soil analysis includes at least one of the following tests: Nitrogen-Nitrate content, pH level, and electrical conductivity. The tests must contain time data and results must be included in the PLAR. The payload must not extend more than a quarter inch from a six-inch maximum diameter airframe, and autonomous sampling and testing can be remotely initiated. The payload must also contain 4 STEMnauts enclosed from the atmosphere and within the overall Habitat for Agricultural Utilization Study (HAUS) during flight and landing. These STEMnauts model real human passengers and thus their safety warrants critical design consideration. An additional experiment can be optionally tested and incorporated into the payload barring approval from NASA for safety. Energetics are only allowed for in-flight operations, and the rocket will need to be registered with the FAA.

After closely reviewing NASA requirements the team formulated Table 2 which addressed each design requirement with a corresponding physical constraint.

Table 2: USLI Payload Requirements versus the Physical Impact on the Payload Design.

Design Requirement	Physical Implication
Collect at least 50 mL soil	Soil collection mechanism must be designed that can properly mine Huntsville soil under a variety of weather conditions.
Analyze soil with one of the following tests: Nitrogen-Nitrate Content, pH level, electrical conductivity	Soil sensor must be incorporated into payload and it may require water to function.
STEMnauts must be enclosed from the atmosphere within the payload housing	Payload containing collection mechanism must be extended sufficiently to accommodate STEMnauts in a separate section.
STEMnaut housing should contain accommodations sufficient for an extended excursion	The HAUS should contain sleeping, eating, exercising, and living quarters.
Energetics are only approved for in-flight operations	Energetics are not permitted for on-the-ground decoupling or soil collection.
Initiation of payload operations must be either remotely initiated or fully autonomous	Operation of soil collection mechanism cannot be entirely remote controlled.

4.1.2 General CONOPS

The first stage of rocket launch is takeoff. The vehicle lifts off under a Cesaroni L2375 motor and ascends to a target apogee of 4,800 ft AGL. Apogee is regulated by the apogee control system

(ACS), which monitors ascent state and commands a deployable drag device when the predictor indicates an overshoot relative to the 4,800-ft target. This closed-loop action trims ascent energy so the vehicle reaches the setpoint and transitions to drogue deployment at apogee. Detailed trajectory modeling, ACS logic, and performance predictions are provided in 4.2.4.

After drogue-assisted free fall, the main parachute will deploy at 550 ft, decoupling the entire air-frame into nose cone, body, and aft payload with fins. During decoupling, four spring-loaded legs in the aft section will deploy once separated from the body. The rocket will fall to the Huntsville, AL soil with a landing energy of no more than 75 ft-lbs according to NASA SLI Recovery System requirement 3.2.

Once on the surface and propped on its side by the four legs and four fins, the payload will be autonomously activated by an on-board STM32F466RE microcontroller. The rotation of the shovel shall shear off a thin outer layer of 3D printed material thus exposing the shovel and soil collection chamber. Continuous motor activation will gradually open the shovel outward at a rate of about 0.92 in/min as the payload bay rotates at 6 rpm. These measurements were taken directly from the CAD model in SolidWorks. Release of the shovel will be governed by a gradual unwinding of Kevlar rope around a central non-rotating shaft as the payload bay rotates. A 23.8 Nm maximum torque motor will drive a shaft housing the Kevlar rope and shovel-container assembly. Once the payload bay rotates a total of 48 times and takes 20 passes of soil, a syringe containing 100 mL water will be passively actuated, thus thoroughly dampening the collected soil. A robust soil sensor ordered online will be mounted to the soil collection chamber and it will interface wirelessly with a VADL computer, sending live soil analysis results to the team.

The general CONOPS mission sequence can be viewed in Figure 25.

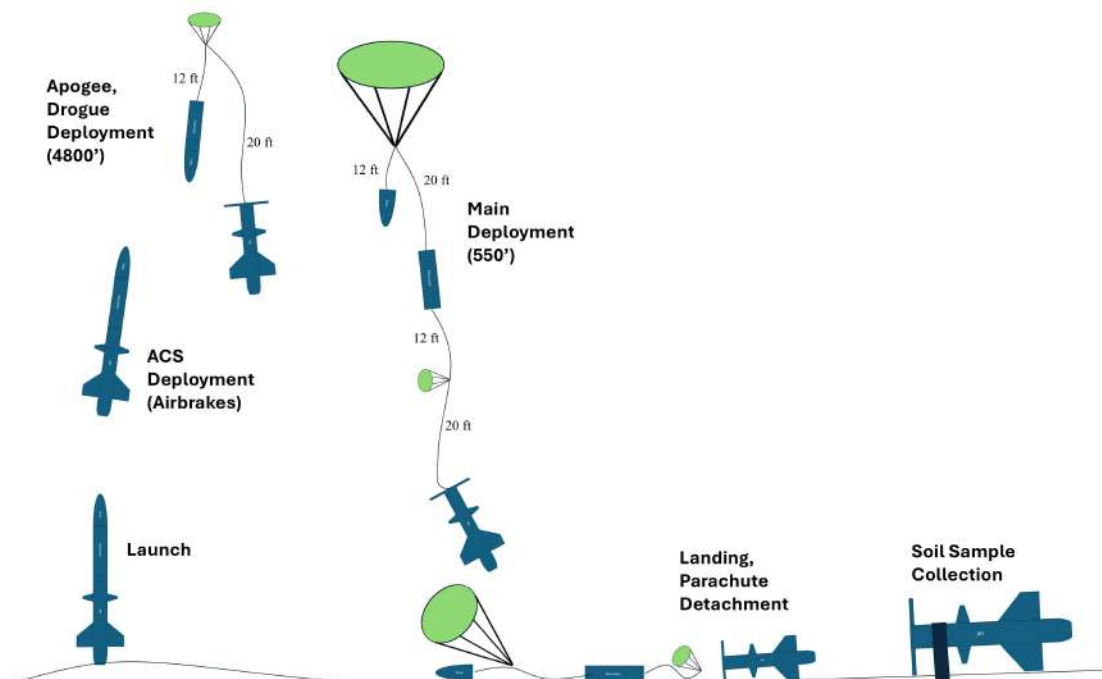


Figure 11: General CONOPS schematic for flight mission sequence.

4.2 Vehicle Design

4.2.1 Rocket Layout and Dimensions

Introduction / Preliminary Design Decisions

NASA's Student Launch challenges teams to design, build, and fly a high-power rocket with a research payload under defined safety and performance constraints. The vehicle described here is the current, proposal-phase concept: a dual-deployment, L-class launch vehicle configured around constraints imposed by the HAUS payload. The layout and dimensions that follow represent the as-modeled baseline used for initial simulations and recovery sizing; they are preliminary and will be refined as the project progresses.

Payload-vehicle integration drives the architecture. Because the rotating-shovel payload (described in Section 4.2) requires a stable base and ground contact, we evaluated two configurations: Payload Fore (legs fore and aft) and Payload Aft (tail-mounted payload with passive legs). Both concepts were sized to maintain safe static stability, meet the 4–6 kft apogee window, and minimize landing kinetic energy under the main parachute.

The initial OpenRocket models used to study these two cases are shown below.

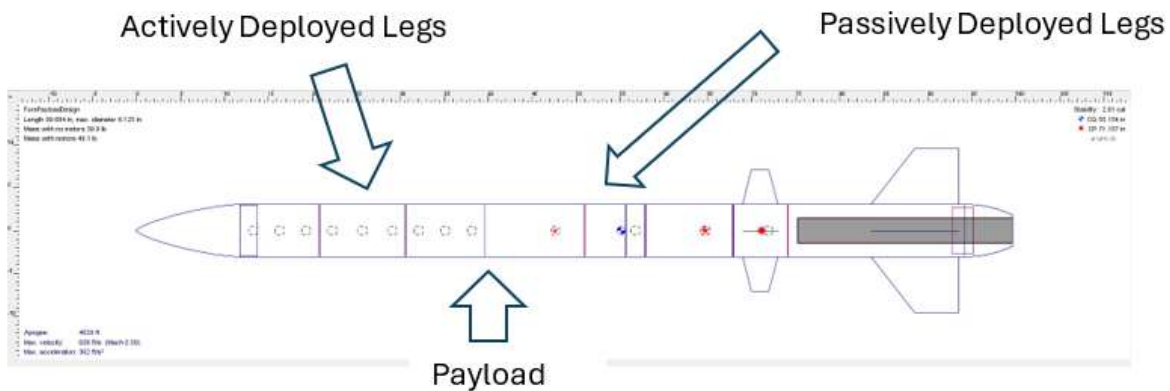


Figure 12: Preliminary OpenRocket model — Payload Fore configuration (payload forward; dual leg sets).

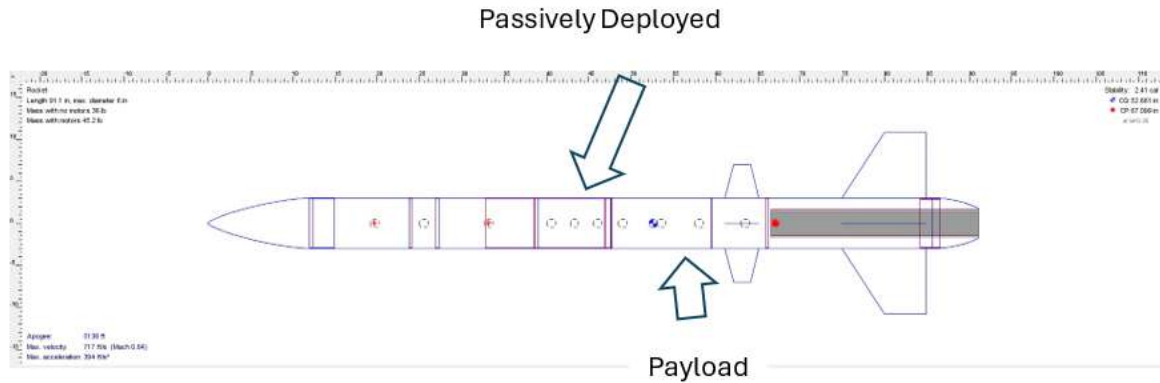


Figure 13: Preliminary OpenRocket model — Payload Aft configuration (tail-mounted payload; single passive leg set).

Results for these two preliminary models are shown in Table 3.

Table 3: First-concept trade between aft and fore payload configurations.

Payload Location	SSM (Cal)	Apogee (ft)	Landing Velocity (ft/s)	Wet Mass (lb)	Max Landing Section Mass (lb)	Max Landing KE (ft-lbf)	Requires Active Leg Actuation
Fore	2.61	4629	14.6	49.1	26.6	88.1	YES
Aft	2.41	5138	13.8	45.2	27.4	77.6	NO

Key points driving the decision between fore and aft designs:

- **Payload Aft**
 - Lighter overall mass and lower maximum landing kinetic energy
 - Single leg set that deploys passively with the drogue (reduced mechanisms and risk)
 - Tail fins act as secondary ground supports, providing a stable base for payload actuation
- **Payload Fore**
 - Requires two actuated leg sets, including one not co-located with the recovery bay
 - Increases mass, part count, and deployment risk

Both preliminary configurations exceed the ≤ 75 ft-lbf landing-energy limit with a 10 ft main chute. The most effective path to compliance is reducing landing velocity via main-canopy sizing and deploy altitude; additionally, mass reduction provides a secondary benefit.

The Payload Aft configuration was selected as the controlling design because it provides lower kinetic energy, lighter mass, and only requires one passively deployed leg set.

Current Vehicle Design

After selecting the Payload Aft configuration, the vehicle was re-designed end to end. A 12 ft parachute was adopted to bring landing kinetic energy within program limits. Additionally, as the payload and other subsystems matured, updated massing and dimensioning were incorporated to produce a higher-fidelity model. This refined Aft configuration with current masses, geometry, and recovery points now serves as the controlling vehicle design for this proposal. The figures below present the as-modeled vehicle layout and principal dimensions that define this proposal baseline.

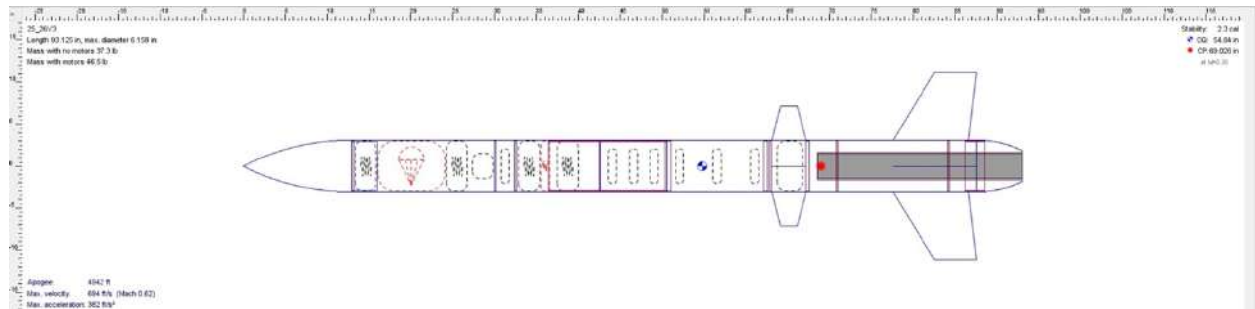


Figure 14: OpenRocket model of the proposal baseline.

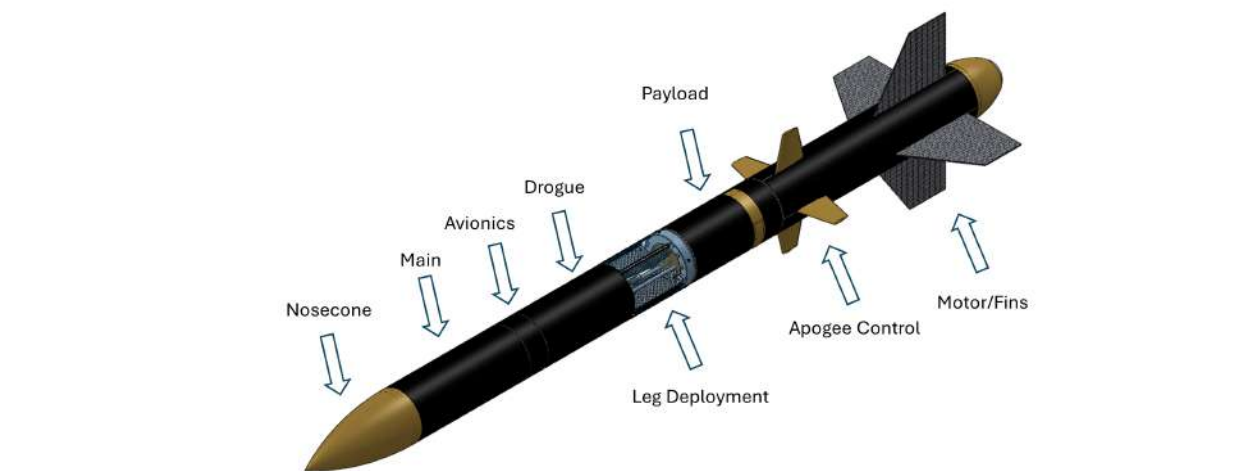


Figure 15: CAD assembly of the as-modeled vehicle layout and principal dimensions.

Table 3 summarizes the as-modeled principal dimensions and mass properties used for trajectory and recovery sizing. Values reflect the current proposal baseline and will be refined as the design matures.

Table 4: Principal dimensions and mass properties (proposal baseline).

SSM (Cal)	Overall Length (in)	Body Tube ID (in)	Nosecone Length and Profile
2.3	93.1	6.0	13", Tangent Ogive

Mass properties drive the CG/CP relationship, vehicle performance, and landing energy. To make weight drivers visible early, we track mass by subsystem and by vehicle section. The charts below show the proposal-phase distribution for liftoff (with motor) and dry (without motor); these allocations set targets for weight control, ballast limits, and section masses used in kinetic-energy calculations and will be refined with part selections and as-built measurements.

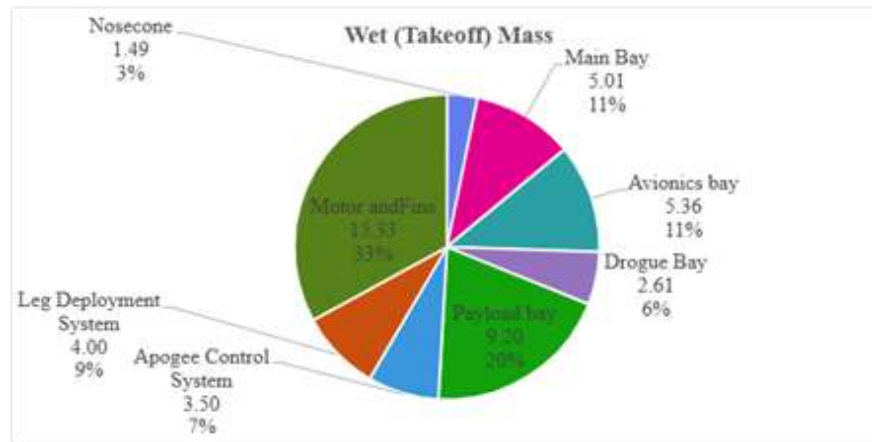


Figure 16: Liftoff mass breakdown by subsystem.

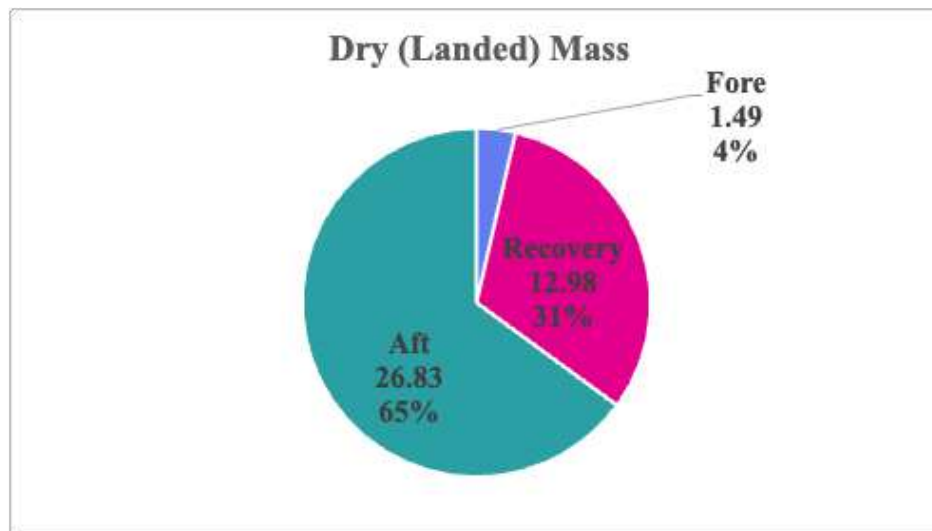


Figure 17: Dry mass breakdown by vehicle section.

4.2.2 Material Selection and Testing Procedures

Material Selection

When choosing a primary material for vehicle construction, two variables were considered: body tube diameter and body tube material. The team evaluated 5.5 in, 6.0 in, and 6.5 in inner diameters and selected 6.0 in ID based on:

- Proven flight success in prior VADL vehicles
- Ready availability of tubes, couplers, and nosecones from vendors and in team stock
- Sufficient packing volume for the tail-mounted payload, avionics, and recovery hardware

In contrast, 5.5 in and 6.5 in options were harder to source consistently at the required lengths/coupler sizes, and each introduced additional integration compromises. After fixing the inner diameter to 6.0 in, the team conducted a material trade study to select the primary airframe material.

Two candidates were compared: (1) a carbon fiber/Blue Tube hybrid layup and (2) fiberglass. The design team conducted a qualitative trade study (scored 1-3, 3 being highest). The summary is provided below.

Table 5: Carbon Fiber/Blue Tube vs. Fiberglass Qualitative Trade Study.

Metric	Carbon Fiber/Blue Tube	Fiberglass
Ease of Construction	1/3	2/3
VADL Flight-Proven	2/3	2/3
Structural Efficiency	3/3	2/3
Impact Toughness	1/3	3/3
Price / Ease of Sourcing	2/3	3/3
Moisture / Temp Stability	2/3	3/3

Fiberglass was chosen due to its availability, predictable fabrication, and consistency with VADL's historical processes.

Fiberglass Testing

To further support the choice of fiberglass body tube construction, the team conducted static load testing to find the strength of sample fiberglass body tube in different configurations, namely axial compression, side-on compression, and bolt tear out. Each load configuration was designed to simulate a principal stress state on the vehicle body. Takeoff compressive forces are simulated in Test 1 (Axial Compression), main parachute deployment load simulated in Test 2 (Bolt Tear Out), and landing impact in Test 3 (Side-On Lateral Compression / Hoop Stress).

Procedure

All tests were performed using a hydraulic load frame. A subscale fiberglass airframe section was tested, and material properties extrapolated from the test results. Length = 7.75 in, OD (do) = 4.02 in, ID (di) = 3.90 in, and bolt hole edge distance (e) = 0.50 in. A single strain gauge was used in each test to record strain. Each test type was conducted once.

Loading Protocol

- **Test 1. Axial Compression:** The tube was mounted vertically and loaded up to 5 kN.
- **Test 2. Bolt Tear Out:** A bolt plate was screwed into the tube with four 10-32s screws that were 0.2 inches in diameter and loaded in direct tension until complete tear-out occurred.

- **Test 3. Hoop Stress:** Loading was applied circumferentially and was taken to yield.

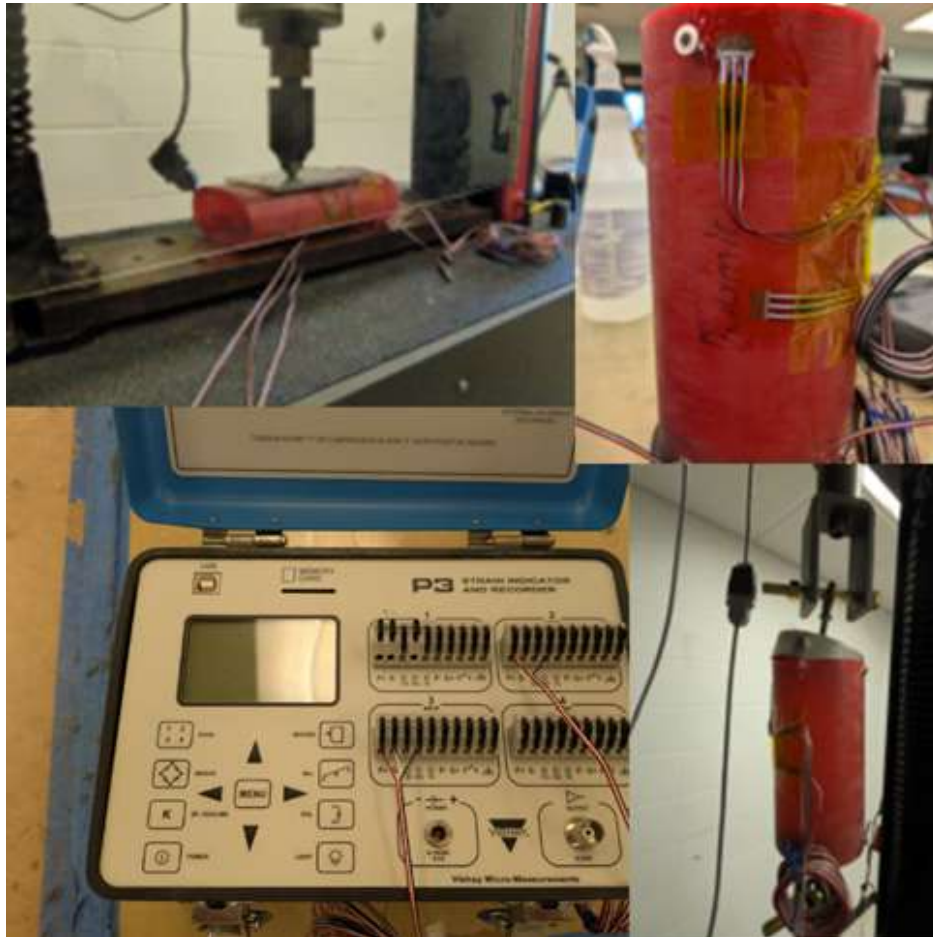


Figure 18: Fiberglass load testing procedure. Top left, axial compression (test 1). Top right, hoop stress (test 2). Bottom right, bolt tear out (Test 2). Bottom left, strain gauge amplifier input.

Results

Test 1. Axial Compression: Young's Modulus calculated at $E=1278$ MPa or 1.28 GPa.

Test 2. Bolt Tear Out: - $\frac{1}{4}$ inch hole egging and eventual tear out at 55MPa.

Test 3. Hoop Stress: Strain Rossette Saturation at 10,000 microstrain (14.8 MPa).

4.2.3 Potential Fabrication Methods

Fiberglass Airframe

VADL plans to use pre-rolled fiberglass body tubes and couplers for its main airframe construction, due to the affordability, time savings, and dimensioning/stability reliability inherent with mass-manufactured composite tubing. When cutting this tube to length, the team plans to use a 2-stage process to ensure square, true edges on each body and coupler section. This process involves an initial oversized cut on the miter saw, followed by line-up turning on the lathe. This will

serve the dual purpose of providing a square edge for each mating section to register against, as well as providing the ability for the team to achieve more accurate section lengths. The team plans to fabricate the rocket nosecone using a 3D-printed female mold for improved surface finish compared to previously used male molds. This process with the new mold design will be tested and improved during subscale fabrication.

Aluminum

Bulkplates will be made mostly of aluminum alloys such as 6061 to withstand the shock tensile loads at parachute deployment. They will be constructed with a combination of traditional machining, where the lathe will be used to create the outer shape and inner main cutout, and the interior asymmetrical holes and features machined on a mill. If some bulk plate designs become complex in an attempt to achieve extra functionality or save weight, VADL also has access to a 4-axis CNC machine. The team does not plan to employ aluminum TIG welding on its bulkplates due to the ability to machine from one block of aluminum (due to the machinability and affordability of aluminum) and because the bulkplates are main failure points during parachute deployment events. Even complete joint penetration welds induce inherent uncertainty into the grain structure of these mission-critical components. However, TIG welding may be employed on vehicle subsystems (such as the Leg Deployment System) where weldments are not loaded during deployment.

Subscale

Prior to the full-scale flight, the team plans to fly a subscale version of our vehicle in order to test flight mechanics, flight logic, and the Apogee Control System. The diameter of this subscale will not exceed 4" and the length will not exceed 68", in accordance with USLI Handbook Requirement 2.16. The team will also use this opportunity to test fiberglass fabrication techniques, and will diligently document the subscale construction process in order to identify and implement improvements for the construction of the full-scale vehicle.

4.2.4 Flight Predictions and Analysis

This section quantifies expected flight performance for the preliminary full-scale vehicle defined in 4.2.1. Using OpenRocket, RockSim, and our custom MATLAB-based Declarative Rocket Simulation Software (DRSS) framework, we predict ascent and descent behavior to inform motor selection, recovery sizing and launch-operations planning. The key objectives were:

- Estimate apogee, descent rates, landing kinetic energy, and total descent time (NASA 2.1, 3.2, 3.11)
- Verify liftoff thrust-to-weight and rail-exit speed (NASA 2.12, 2.14)
- Confirm stability margins across the flight envelope (NASA 2.11)

Results will be continuously refined as the payload, vehicle, and recovery designs mature, with models re-baselined after each configuration change.

Simulation Inputs – Overview

The vehicle geometry and mass properties are defined in 4.2.1; here we list only the simulation-specific parameters that govern the trajectories. Propulsion (4.2.6) and recovery (4.2.5) are summarized because they affect apogee, rail-exit, descent rate, and drift. Results reflect nominal, deterministic runs: a zero-wind case and a single design-wind case. In DRSS, ascent drag is modeled with $C_d = 0.70$ (constant) for proposal.

Table 6: Launch parameters and recovery configuration.

Parameter	Value
Launch Angle (degrees)	0
Launch Rail Length (ft)	12
Wind Model (m/s)	0
Launch Elevation (ft)	0
Motor (brand, designation, total impulse)	Cesaroni, L2375-P, 4,864 Ns
Drogue parachute (diameter (in), C_d)	24, 1.6
Main parachute (diameter (in), C_d)	144, 2.2
Deployment altitudes (drogue, main, ft)	Apogee, 550

Ascent drag coefficient, $C_d = 0.70$, taken from OpenRocket’s auto C_d for the as-modeled geometry.

Simulation results

Figures 19 and 20 show altitude–time profiles for the no-wind nominal case from two tools: Figure 19 (OpenRocket) and Figure 20 (DRSS). Both use the same geometry and mass configuration from 4.1.1 and the nominal simulation parameters listed above. These deterministic runs provide an early read on apogee, rail-exit performance, and descent behavior.

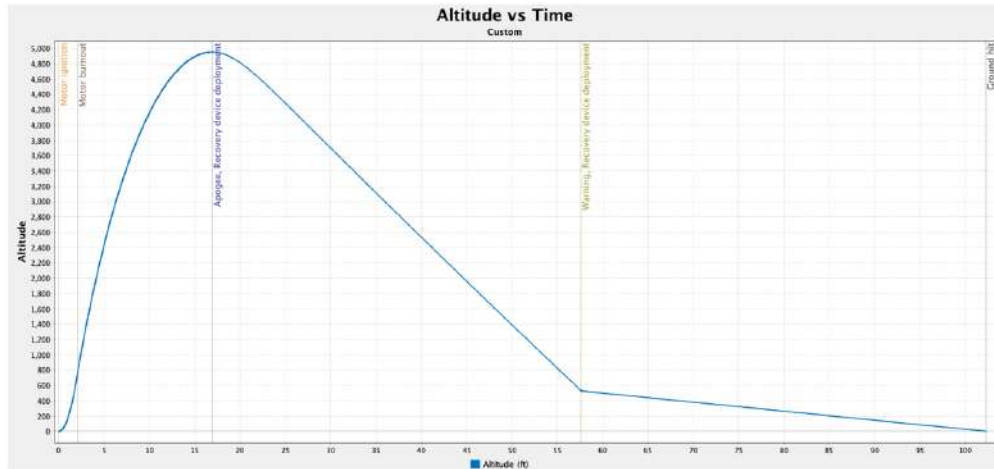


Figure 19: Altitude–time profile with no Apogee Control System (ACS) deployed (OpenRocket, no-wind). The ACS will be discussed in much greater detail in 4.6.

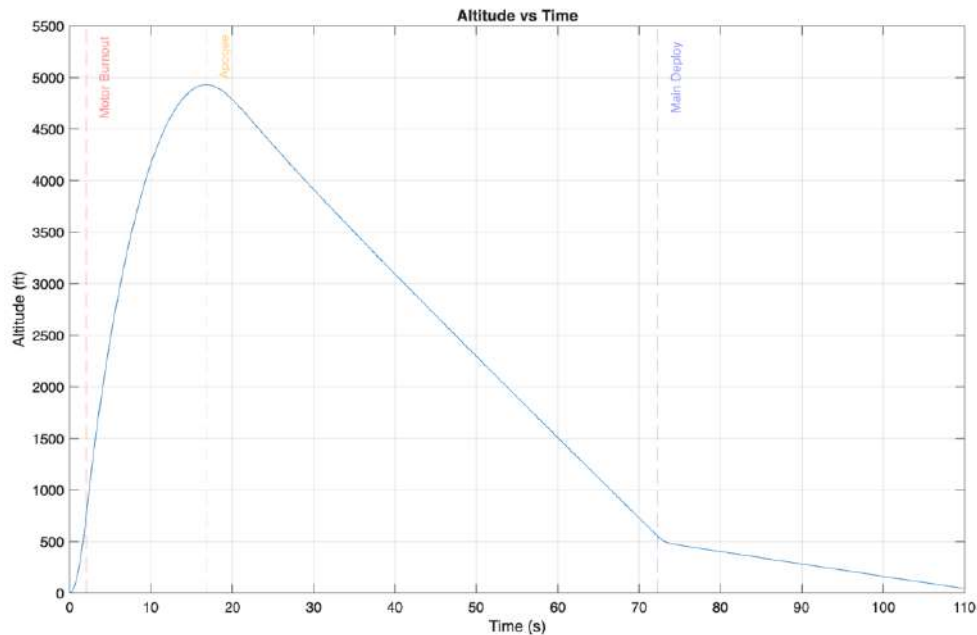


Figure 20: Altitude–time profile with no ACS deployed (DRSS, no-wind).

Simulations incorporating the ACS system yield a controlled apogee of ~4800 ft (Figure 21).

Requirement compliance:

- **Apogee:** All predicted apogees fall within NASA Requirement 2.1’s altitude window of 4,000–6,000 ft.
- **Rail-exit velocity:** All tools predict rail-exit speed ≥ 52 ft/s, satisfying Requirement 2.14.
- **Landing kinetic energy:** With tail-section landing velocities of 11.8–11.9 ft/s, the current estimates yield 58–59 ft-lbf based on the tail section mass of 26.8 lb—below the program limit of ≤ 75 ft-lbf per free section.

Table 7: Key nominal metrics (zero-wind, no ACS deployment).

Metric	OpenRocket	RockSim	DRSS
Apogee (ft)	4952	5011	4929
Static Stability Margin at Liftoff (cal)	2.3	2.38	2.4
LRE Velocity (ft/s)	91.1	90.2	94.6
Landing Velocity (ft/s)	11.8	11.0	11.9
Tail Section Landing KE (ft-lbf)	58.0	50.4	59.0
Total Descent Time (s)	84.7	82.3	96.5

- **Descent time:** OpenRocket and RockSim satisfy Requirement 3.11 (≤ 90 s). The current DRSS nominal slightly exceeds 90 s; as the design matures, drogue sizing will be adjusted to ensure compliance. Additionally, by implementing the ACS and lowering the apogee, the descent time will also decrease.
- **Drift:** Using the planning estimate $d \approx t_d w$, even with the DRSS worst-case descent time of $t_d = 96.5$ s and a 15 mph wind ($w = 22$ ft/s), the drift is $d = 2123$ ft—within the 2,500-ft recovery radius specified by Requirement 3.10.

Apogee Prediction – ACS Deployed

To quantify apogee with apogee control engaged, the ACS was incorporated into DRSS as an event-driven drag-augmentation model representing the deployable drag plate. In this closed-loop case, the ACS targets an apogee of 4,800 ft AGL. DRSS implements the same Runge–Kutta fourth-order guidance and timing logic planned for the flight unit to command deployment when the predictor indicates an impending overshoot of the 4,800-ft target. At deployment, the ascent drag coefficient is stepped from the baseline to a higher effective value reflecting the plate’s planform and interference drag. Under these assumptions, the simulation reaches an apogee of 4,816 ft. All other inputs and initial conditions (motor file, mass properties, pre-deployment drag, and recovery settings) match 4.2.4. The corresponding adjusted altitude–time profile is shown below.

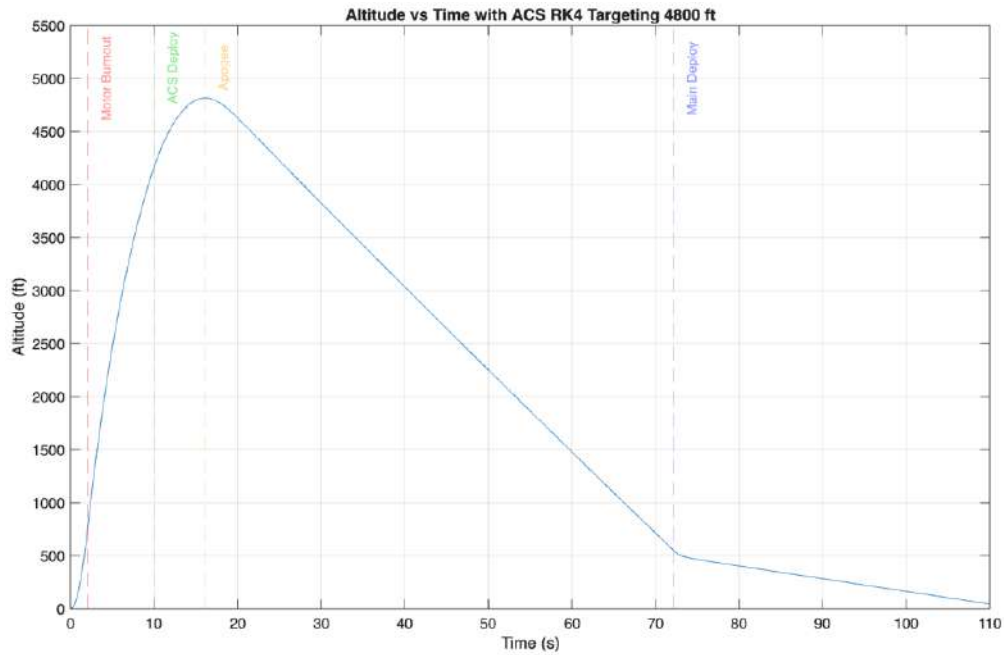


Figure 21: Altitude–time profile with ACS deployed (DRSS, no-wind).

Based on the current final configuration, including the modeled ACS deployment, the predicted apogee is **4,800 ft**; additional simulations will be run to optimize the deployment altitude and quantify the drag-augmentation effects.

4.2.5 Recovery System Design

Safe, reliable recovery is a primary design driver because it governs descent time, landing kinetic energy, and public/crew safety. The baseline is dual-deployment: a Fruity Chutes Elliptical drogue at apogee to stabilize the vehicle and control descent energy, followed by a Fruity Chutes Iris Ultra main at 550 ft AGL to achieve a predictable, low landing rate. Canopy diameters are selected to balance three objectives: maintaining total descent time within the ≤ 90 s requirement, limiting landing velocity so each free section remains ≤ 75 ft-lbf kinetic energy, and managing opening loads at main deployment.

With these objectives in mind, we conducted a comparison of candidate canopy sizes. The values below were generated in OpenRocket using the 4.2.1 configuration. Green cells indicate compliance with program limits; red cells indicate a value requiring adjustment.

Table 8: Recovery canopy sizing trade (OpenRocket; nominal no-wind).

Config	Main Ø (ft)	Drogue Ø (in)	Descent time (s)	Ground-hit vel. (ft/s)	Max accel (g)
1	10.0	24.0	90.1	14.0	34.4
2	12.0	24.0	97.8	11.8	49.5
3 (chosen)	12.0	18.0	84.7	11.8	80.1

We have currently selected Config 3 (12-ft main, 18-in drogue) because it satisfies the descent-time requirement (84.7 s) while maintaining an acceptable landing velocity (11.8 ft/s), keeping section landing energy below the ≤ 75 ft-lbf limit. Although OpenRocket reports a high peak acceleration for Config 3 of 80.1 G, that value is driven by the tool’s instantaneous-inflation assumption for the main; measured openings typically occur over a finite fill distance and produce lower peak loads. We will refine these predictions with staged-inflation modeling in DRSS and with instrumented ground-ejection and flight tests before finalizing canopy sizes and deployment altitude. These refined predictions will be utilized in further design and will be discussed in further design reports.

Shock-Cord Sizing

Shock cords connect each parachute to its anchoring hardware and absorb the transient loads at deployment. Cord sizes are derived from first-principles calculations using the simulated deployment conditions: deployment altitude, vehicle speed at the event, canopy model and drag, and the assumed inflation behavior. For main deployment jerk we apply a conservative instantaneous-opening model to establish the required cord type and length. Key considerations include maintaining sufficient separation between sections while keeping total shock-cord length as low as practical to reduce mass and to minimize jerk at deployment so the cords take load only after the canopy is largely unfurled. Because the payload is located in the tail section, the preferred touchdown orientation is tail-first; rigging lengths and attachment geometry are chosen to promote this behavior. As a general rule, the effective cord length is targeted at approximately four times the length of the mated section to prevent contact during descent. Final lengths will be refined through ground-ejection testing and flight data to preserve descent-time and landing-energy margins. Preliminary selections use one-half-inch Kevlar tubular webbing. Figure 22 illustrates the current layout and nominal lengths for both deployment modes.

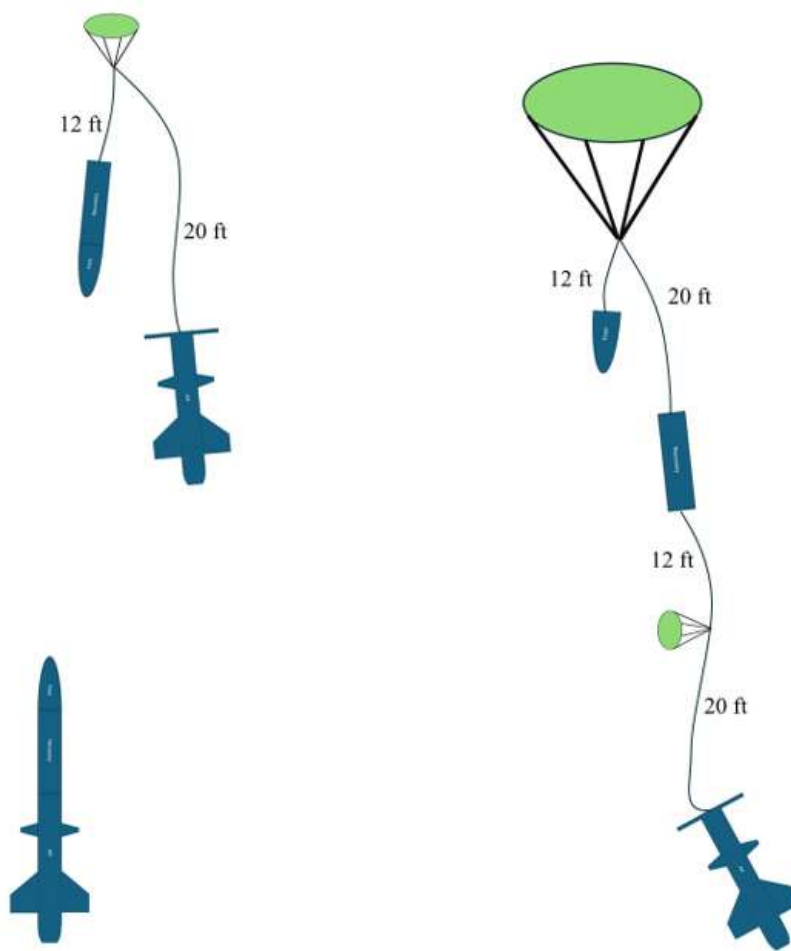


Figure 22: Preliminary shock-cord layout and nominal lengths for dual-deployment recovery.

4.2.6 Motor Selection

To achieve the 4800 ft apogee target, the Cesaroni L2375 motor has been identified as the primary motor, providing sufficient thrust to enable the vehicle to reach this altitude with apogee control system (ACS) correction. Table 7 summarizes the performance of the preliminary design without ACS actuation using the L2375 motor. Should testing and simulation indicate that the vehicle mass is overestimated, the total flight time exceeds the 90-second requirement, or the landing kinetic energy surpasses the 75 ft-lbf safety threshold, the AeroTech L1940X has been identified as an alternate motor that mitigates these risks. Table 9 presents the dimensions and propulsion data for both motors. With the L1940X configuration, the expected apogee decreases to approximately 4100 ft with ACS deployment.

Table 9: AeroTech L1940X and Cesaroni L2375 motor dimensions and data

Parameter	L1940X	L2375
Diameter (in)	2.95	2.95
Length (in)	22.0	24.4
Total Impulse (N·s)	4329	4864
Burn Time (s)	2.2	1.9
Peak Thrust (N)	2319	2798
Delay Time	Plugged with smoke	No delay (ejection)
Propellant Mass (lb)	4.02	5.12
Loaded Mass (lb)	8.50	9.17

4.3 Payload Design

4.3.1 Payload Design Evolution and Selection

Initial design ideas for this year's design challenge centered around an auger payload. The team created drawings and rough CAD models of a jettisoned payload which could drill into the ground, collect soil, retract, and finally spin at a rapid rpm to release the soil into a catchment bin for testing. A separate auger design was proposed that repositioning to orient the auger downward after the airframe had landed on its side (named the auger non-jettisoned lander). The main problem encountered with this design was the multiple actuators required for plunging and rotating the auger towards the ground, and potentially a third actuator to add moisture to the soil.

Another leader in the design brainstorming process was a rotating bucket chain (rotating buckets) design with several buckets on a rotating belt to steadily collect soil and still collect the soil even if a number of buckets failed. The main issue with the rotating bucket design was multiple moving parts and failure points in the bucket assembly as well as the added complexity of another actuator to add moisture to the soil. A third group of designs was centered around a hole punch syringe style sampler or a coring bit to drill into the ground with the sensor inside the part biting into the soil. In the hole punch design the sampler would push into the ground using a spring loaded mechanism to encapsulate the soil, while in a coring bit the soil would get trapped between the walls of the bit during penetration. These designs did not incorporate a mechanism for adding water to the soil which we deemed pivotal for accurate soil testing results due to the required threshold for moisture on our sensors.

General issues that were present in all the designs were complexity in design and number of actuators. However, a design that we had previously overlooked addressed both of these issues. This design was the rotating shovel idea which incorporated only one actuator for the task of shovel extension, payload re-orientation, soil collection, soil testing, and shovel retraction. The shovel incorporates its rotation from a central motor to rotate the payload bay which in turn unwinds Kevlar rope around its central non-rotating shaft to passively extend the shovel. As the shovel extends, it begins taking progressively larger passes of soil and dumping that soil into a soil collection chamber. The rotation of the payload section also forces a syringe to push water into the collection chamber by rotating around a central threaded rod. A soil testing sensor will be mounted to the collection chamber and constantly interface with a VADL computer taking measurements so that it can record soil data once the soil is collected and moisturized. A single motor drives the entire payload actuation; as it rotates, it uses its built-in encoder to determine the appropriate stopping point for soil collection. Once the shovel has completed 48 passes, the motor will reverse, spinning the payload in the opposite direction and retracting the shovel.

After analysis of several different designs as shown in Table 1 below, the team determined that the rotating shovel design was actually more practical, likely to succeed, and elegant.

Table 10: Comparison Matrix of Different Payload Designs. Specific details and models relating to each design are mentioned in the subsequent paragraph.

Payload Design	Stability	Actuation Simplicity	Sample Chamber	Form Factor	Failure Points	Innovation
Rotating Shovel	9/10	8/10	9/10	8/10	8/10	42/50
Coring Bit	8/10	4/10	8/10	6/10	7/10	33/50
Hole punch sampler	7/10	7/10	7/10	5/10	7/10	33/50
Horizontal Scraper	6/10	6/10	8/10	4/10	8/10	32/50
Rotating Buckets	7/10	3/10	8/10	4/10	9/10	31/50
Auger (Jettisoned Lander)	2/10	5/10	5/10	4/10	6/10	22/50
Auger (Non-Jettisoned Lander)	9/10	4/10	6/10	5/10	3/10	25/50

This year the team wanted to prioritize reliability, while keeping innovation as a close secondary goal. It is the view of VADL that new designs should not only meet all the USLI requirements but should also introduce a new concept that can further our understanding of possible solutions. A good example of implementing this philosophy was the rule-out of the auger idea. Mainly, it was deleted because of the extremely high torque requirements for actuation, and the presence of a difficult to manufacture anchoring base. Additionally, the auger was not highly innovative. The combination of the potential un-reliability with the lack of innovation in this design led to its removal from consideration.

Believing that there was a way to improve some of the flaws of the auger, the team tested using a coring bit on soil as seen in Figure 23. A coring bit was believed to have lower torque upon actuation, and a intrinsic ‘soil-catcher’, as the soil would get caught in the bit’s cavity during actuation. During testing, it was effective at mining soil; however, as mentioned previously there must be multiple actuators required to implement such a design, giving it excess points of failure that do not outweigh the benefits of choosing it.



Figure 23: Testing of coring bit on Nashville, TN soil. Soil assumed to be similar to the conditions at competition in Huntsville, AL.

Another contending design was tested to ensure that development of the shovel should be pursued. A horizontal scraping shovel design as mentioned in Table 10 was deployed physically. Interestingly, the torque requirements on the shovel actuation were relatively low, as the test was done by pushing the shovel in nearly horizontal to the ground plane and about 0.10 inches below the surface. However, it was ultimately ruled out due to high number of actuators; at the very least there would be an actuator for translation, rotation of the scraper, and water dispensing. The results of the soil it collected can be viewed in Figure 24.



Figure 24: Horizontal scraper design testing.

While exploring sampling mechanism, the team found that water was a requirement for soil testing due to the hardware specifically being designed for a water-to-soil ratio of at least 1:4, based on the manufacturers spec on the sensor. This moisture content is not feasible to expect in soil, especially if the soil in Huntsville has not experienced rainfall in several days. The added requirement of some sort of water injection potentially required an additional actuator which caused the team to prioritize simple actuation designs like the rotating shovel design even more so.

4.3.2 Shovel CONOPS

During touchdown, the rocket will fall in a horizontal orientation resting on its aft fins and landing legs. Once secured, the control software will command the motor to begin rotation. This rotary motion deploys and actuates the shovel. The shovel performs soil passes while the rotation simultaneously triggers the linear actuation of a syringe system to inject 80 mL of water into the soil chamber. Soil and water are achieved by mixing through the combined shovel motion and chamber rotation. This system will operate for 48 rotations over 8 minutes, nominally collecting a range of 100 to 150 mL of soil. After completion, rotation ceases, allowing the soil-water mixture to stabilize, and the onboard sensors within the testing chamber record data for an additional 7 minutes. The mission CONOPs including timestamps is illustrated in Figure 25.

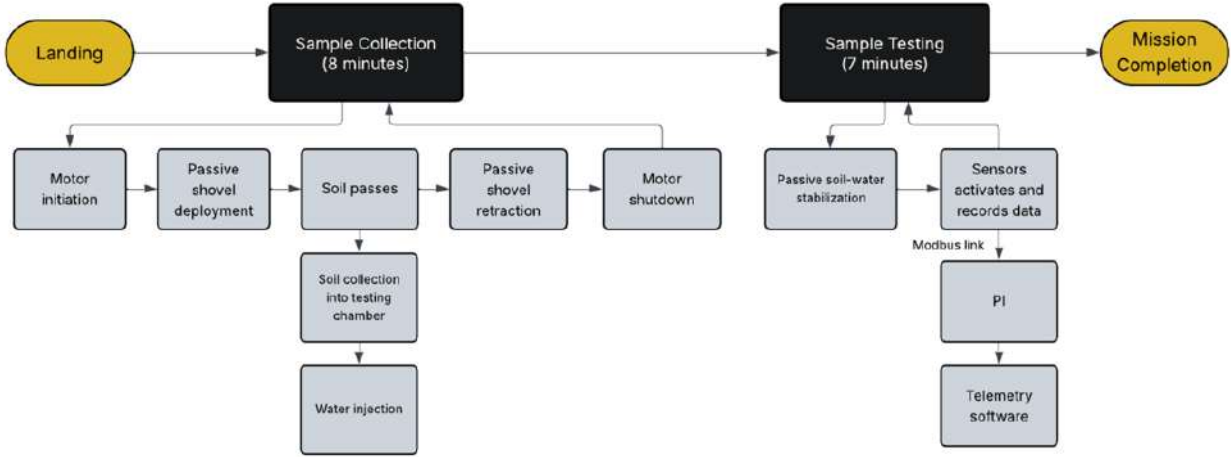


Figure 25: Huntsville surface mission concept of operations for soil sampling and testing.

4.3.3 Hand Calculations

Shovel Load Calculation

The payload team did the following hand calculations to validate the hypothesis that a rotating shovel mechanism could in fact penetrate and dig at least 50 ml of Huntsville, AL soil. These calculations are focused on the *major technical challenges* that are projected to come with the chosen payload configuration. The first assumption made was that to account for residue lost in the assembly, around 90 ml of soil would actually need to be collected. The next assumption made was that the shovel's surface area would be around 25.4 mm × 30.6 mm (Width × Length as measured in SolidWorks). The shovel was assumed to take 3 mm passes of soil for 24 passes for a total depth of 72 mm.

Undrained shear stress for the soil at Huntsville (c_u) was taken to be between 25–150 kPa [1]. For undrained shallow soil, a bearing capacity factor for cohesion (N_c) was taken to be 5.14 [2]. Terzaghi theory was used to find resistance on the face of the shovel and an adhesion factor (α) was assumed to be 0.45 [3].

Equation 1 describes the total force experienced on the shovel face according to a simplified version of Terzaghi Theory.

$$F_{\text{total}} = F_{\text{face}} + F_{\text{side}} \quad (1)$$

with

$$F_{\text{face}} = q \cdot A_{\text{face}} = \frac{N_c \cdot S_u \cdot \text{width} \cdot \text{depth}}{\# \text{ of passes}} \quad (2)$$

Equation 2 describes the force experienced on the face of the shovel, and it can be calculated using the values mentioned previously to solve for minimum and maximum force on the face of the shovel in **Equations 3 and 4**.

$$F_{\text{face,min}} = \frac{5.14 \cdot 25,000 \text{ Pa} \cdot 0.0254 \text{ m} \cdot 0.072 \text{ m}}{24 \text{ passes}} = 9.79 \text{ N} \quad (3)$$

$$F_{\text{face,max}} = \frac{5.14 \cdot 150,000 \text{ Pa} \cdot 0.0254 \text{ m} \cdot 0.072 \text{ m}}{24 \text{ passes}} = 58.75 \text{ N} \quad (4)$$

Equation 5 describes the force experienced on the side of the shovel.

$$F_{\text{side}} = \frac{\alpha \cdot S_u \cdot 2 \cdot A_{\text{side}}}{\# \text{ of passes}} = \frac{\alpha \cdot S_u \cdot 2 \cdot \text{depth} \cdot \text{length}}{\# \text{ of passes}} \quad (5)$$

Equations 6 and 7 were calculated using the above values to solve **Equation 12** for the maximum and minimum forces experienced on the shovel's side.

$$F_{\text{side,min}} = \frac{0.45 \cdot 25,000 \text{ Pa} \cdot 2 \cdot 0.072 \text{ m} \cdot 0.0306 \text{ m}}{24 \text{ passes}} = 2.07 \text{ N} \quad (6)$$

$$F_{\text{side,max}} = \frac{0.45 \cdot 150,000 \text{ Pa} \cdot 2 \cdot 0.072 \text{ m} \cdot 0.0306 \text{ m}}{24 \text{ passes}} = 12.40 \text{ N} \quad (7)$$

Equations 8 and 9 represent the sum of the forces at a minimum and maximum respectively, referencing **Equation 1**.

$$F_{\text{total,min}} = 11.86 \text{ N} \quad (8)$$

$$F_{\text{total,max}} = 71.14 \text{ N} \quad (9)$$

Equation 10 can be used to solve for the torque requirement on the shovel arm. A shovel arm radius of 14 cm was assumed.

$$\tau = F \cdot r \quad (10)$$

Equations 11 and 12 demonstrate the minimum and maximum torque loads expected on the shovel arm.

$$\tau_{\text{min}} = F_{\text{min}} \cdot r = 26.51 \text{ N} \cdot 0.14 \text{ m} = \mathbf{1.66 \text{ Nm}} \quad (11)$$

$$\tau_{\text{max}} = F_{\text{max}} \cdot r = 159.03 \text{ N} \cdot 0.14 \text{ m} = \mathbf{9.96 \text{ Nm}} \quad (12)$$

With our 23.8 Nm maximum torque motor, we should safely meet this general torque requirement.

Tipping Calculation

To begin estimation for tipping of the payload section under load, first the weight of the tail section was taken as 26.82 lbs (or 119.3 N) referencing Section 4.1.1. The forces and torques assumed to be involved in this calculation were the reaction forces from the legs/fins of the section and the force exerted from the shovel on the ground. The distance from the shovel to the ground was taken to be 5.5 inches or 14 cm. Reference dimensions used can be viewed in the schematic in Figure 26. For the shovel not to tip the payload section Equation 13 must hold true:

$$F_{\max} \cdot h_{cg} < W \cdot b_{\text{width half}} \quad (13)$$

From the previous section, $F_{\max}$ is taken to be 71.14 N and the height of the payload center of gravity above the ground is roughly 15 cm. With supporting fins/legs extending 7.5 inches from the airframe at 45° from vertical, the half base width between the legs becomes 13.5 cm, and solving Equation 13 yields Equation 14:

$$71.14 \text{ N} \cdot 0.15 \text{ m} < 119.3 \text{ N} \cdot 0.135 \text{ m} = 10.67 \text{ Nm} < 16.11 \text{ Nm} \quad (14)$$

Therefore, it can be safely concluded that the payload will not tip even under the maximum expected soil force. This offers a factor of safety of 1.51 which we will continue to bolster in the coming weeks by either decreasing the pass depth, or increasing the fin length. Another advantage of the current shovel design is that even if tipping were to occur under maximal loads, the payload could still continue to dig in different orientations since it undergoes full rotations.

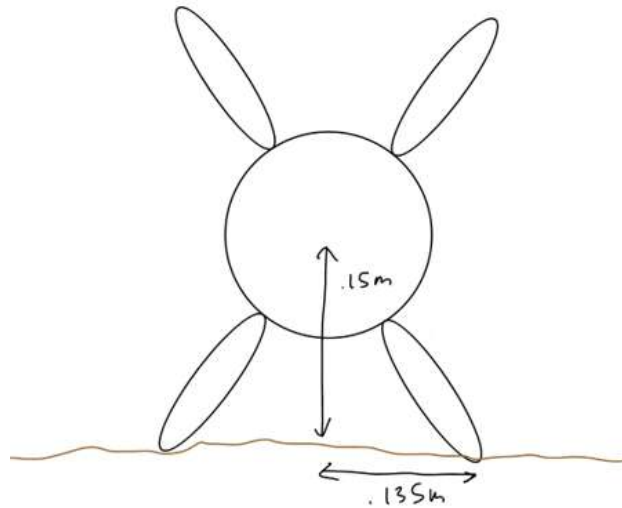


Figure 26: Tipping Calculation Schematic.

Rope Break Calculation

The following calculations were done to understand the strength of the rope used for the opening mechanism. **Figure 4** shows the diagram of the shovel, rope guiding cylinder and force balance used for the calculations.

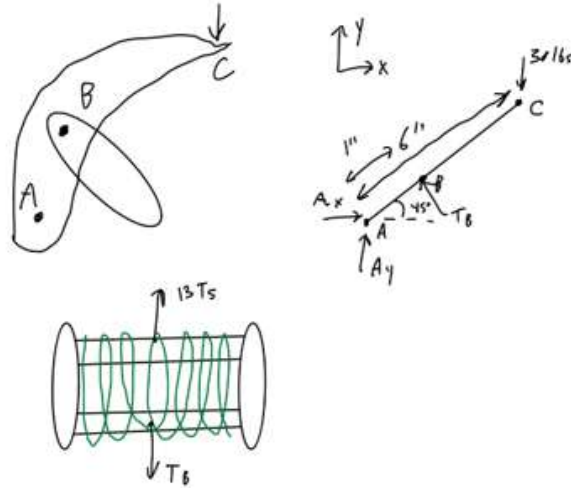


Figure 27: Tipping Calculation Schematic.

Equation 15 was used to find the compressive force (T_B) the shovel arm experiences.

$$\sum M_A = T_B(1 \text{ in}) - 30 \text{ lb} (6 \text{ in} \cdot \sin(45^\circ)) = 0 \quad (15)$$

$$T_B = 127.3 \text{ lbs} \quad (16)$$

Equation 17 was calculated to solve for the tension in the rope.

$$\sum F = 13T_S - T_B = 0 \quad (17)$$

$$T_S = 9.79 \text{ lbs} \quad (18)$$

Equation 12 was used to solve for the safety factor for the rope used for testing with the tensile strength of the rope being 750 lbs according to the manufacturer. The extreme magnitude of this safety factor suggests that a much weaker rope can be used for future testing.

$$SF = \frac{750 \text{ lbs}}{9.79 \text{ lbs}} = 76.6 \quad (19)$$

4.3.4 Mechanical Design

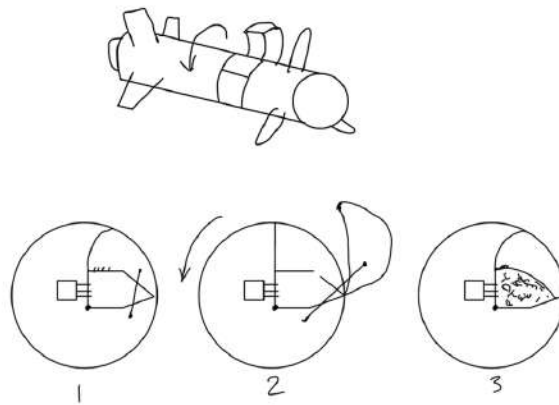


Figure 28: Initial Payload Design.

Figure 28 portrays the initial payload sketch drawn of the rotating shovel design. The sketch features 3 legs that were ostensibly deployed during decoupling and are working in tandem with 4 tail fins to support an excavator-like shovel to emerge from the airframe and begin to take passes at the soil. This design also features a rough idea of a basin with a sensor that is catching the soil flung in from the soil. After 1-2 weeks of further sketches and brainstorming, Figure 28 was created in SolidWorks.

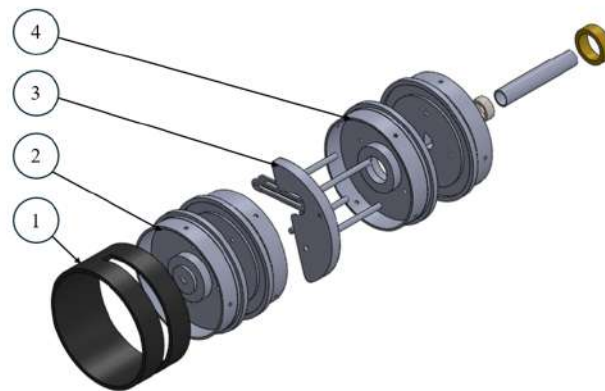


Figure 29: Initial Payload CAD Exploded Assembly. 1 represents the rocket airframe. 2 represents a non-rotating bulk plate. 3 represents the preliminary shovel design. 4 represents a rotating bulk plate.

Figure 29 features a first glance at what a potential shovel could look like. This shovel incorporates a semi-circular profile to sit flush with the airframe and a rail to be adjusted to whatever mechanism the team proposed to let the shovel out. This design also features sets of rotating and non-rotating bulk plates which ensure that only the payload section is rotated. At this point in the design process, the shovel opening mechanism was still largely undecided, and, even today, it remains a point of contention, with plans to be further iterated on for PDR. One potential shortcoming of this design

was the lack of a steep angle and sharp tip to reduce the torque required to dig tough soil. This idea was iterated on with the advent of the second payload prototype in Figure 30.

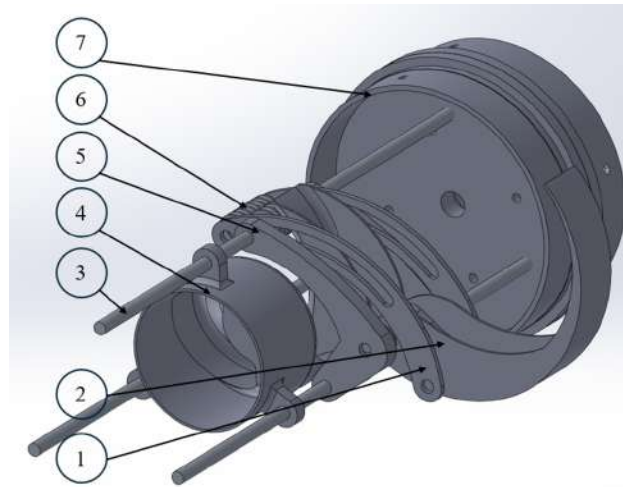


Figure 30: Second CAD model of Payload Assembly. 1 represents a shovel-supporting arm. 2 represents the second shovel design. 3 represents a stabilizing rod. 4 represents the soil-collection chamber. 5 represents an additional supporting arm. 6 represents a cylinder guide for the rope to wrap around. 7 represents the rotating bulk plate system.

The payload design in Figure 30 incorporated a shovel body with a much longer and thinner arm for digging. The hypothesis for this design was that the modified shovel tip would be able to reach further out and take gentler passes due to its sleek, narrow tip. This was the first design to incorporate a mechanism for opening the shovel. This design uses threaded cylinders that match the axial pitch of 2.3 mm rope chosen by the team, so that the rope could be wrapped around the cylinders 13 times. The rotating payload section then unwinds that string by wrapping it around a central non-rotating shaft which slowly and linearly unfurls the shovel. The major shortcoming of this design was in the profile of the shovel. Through real-world testing, the angle of attack of the shovel was not aggressive enough to punch into the ground, so the shovel instead simply pushed its posterior lip against the ground rather than taking passes. This design flaw was iterated on with the third SolidWorks prototype seen in Figure 31. This design (and subsequent designs) maintains the rotating bulk plate system.

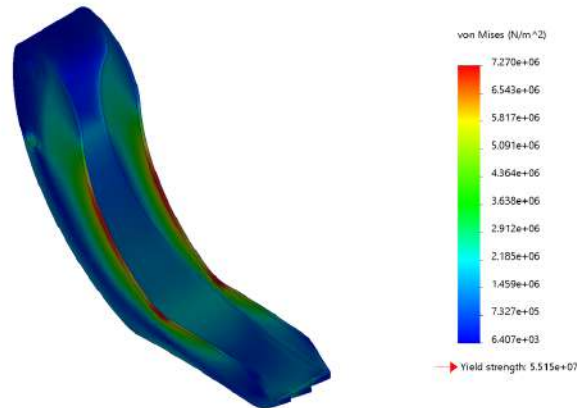


Figure 31: FEA Analysis of Third Shovel Prototype.

The proposed shovel design in Figure 31 features a shovel tip with a much sharper angle of attack toward the ground. This design also mimics the teeth seen in excavator buckets as it was hypothesized that they would allow for better penetration into the soil. The Von Mises stress analysis shown in Figure 31 inspires confidence due to stress concentrated on small areas of the upper sides of the shovel. These areas can be reinforced in later iterations if they prove to be points of failure. A slightly different shovel profile was created that removed the three excavator-like teeth in favor of one single point as seen in Figure 32. The prototype shown in Figure 32 performed similarly to the design proposed in Figure 31; however, the single tooth was felt to be able to puncture the ground with less resistance, so it was kept in later CAD models.



Figure 32: Fourth shovel prototype testing. Soil was collected from nearby yard, and compacted to replicate competition conditions.

Since the prototype seen in Figure 32 has been tested, more emphasis has been placed on the redesign of the shovel opening mechanism. While the Kevlar rope opening system has shown promise in testing, it requires a spring to keep the shovel from buckling outwards with the upward force of soil on its tip when the shovel first touches the soil at the start of the pass. The shovel is strong in tension with the rope but lacks compression strength without the springs which add unconventional sideways loading to the shovel assembly. Another drawback of the Kevlar rope system is that it cannot seamlessly retract the shovel; only when rotating the shovel in the other direction very slowly does it close. Both concerns have been addressed in the latest schematic drawing of a potential new shovel opening mechanism as seen in Figure 33.

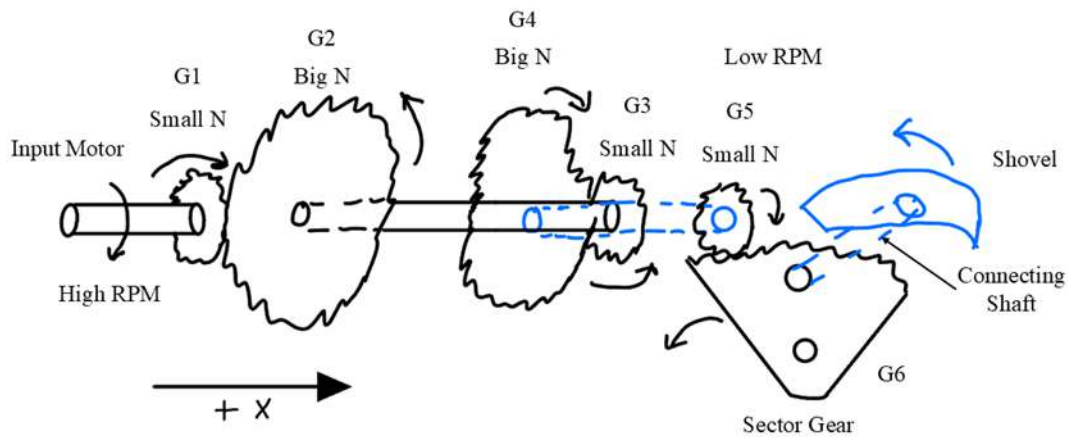


Figure 33: Potential Shovel Opening Mechanism involving Gear Ratios.

The design shown in Figure 33 uses the concept of gear ratios to reduce the relatively high RPM of the driving input motor to an RPM that is suitable for driving a pinion and sector gear. The subdued motion of the pinion gear would work to induce angular displacement of the sector gear which would be attached with a shaft to the shovel, thus angularly dragging it open. This design is very recent and therefore does not have CAD, nor real-world testing to validate it; however, our team will be moving to test it in the coming week.

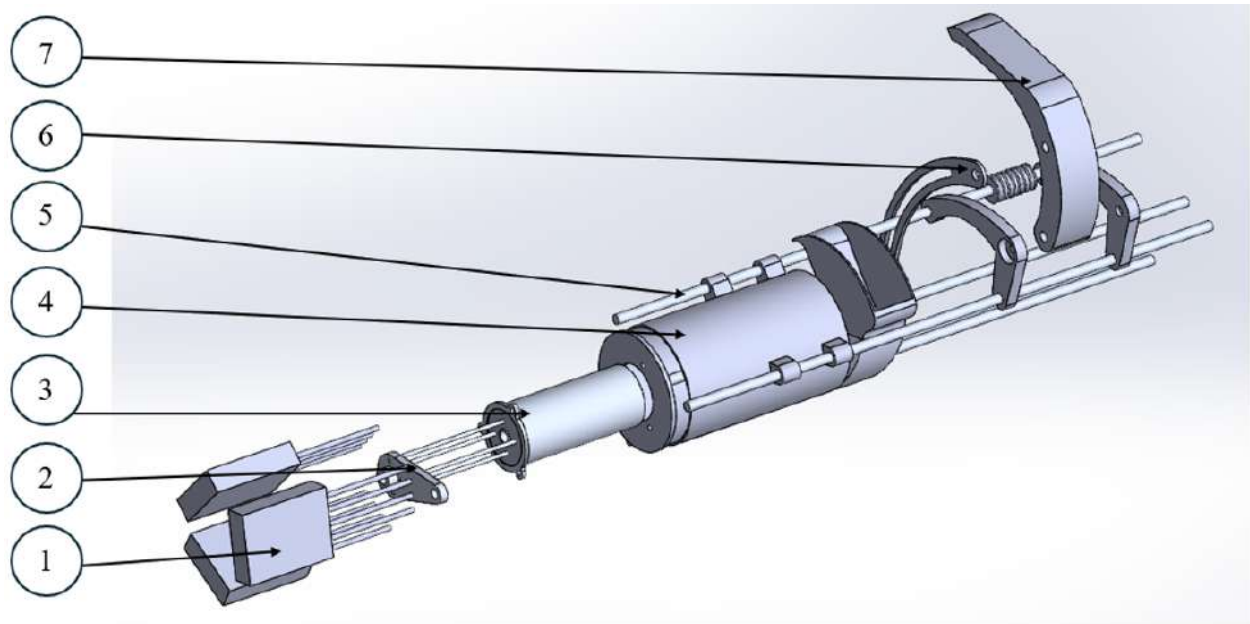


Figure 34: Proposed Design for Syringe Actuation featuring Exploded configuration of shovel, collection chamber, syringe and triplicated sensors. 1 represents the triplicated sensors; 2 represents the part intended to rotate around the threaded rod to actuate the syringe; 3 represents the syringe design; 4 represents the soil-collection chamber; 5 represents the mounting rods; 6 represents the supporting arm of the shovel; 7 represents the latest shovel prototype.

The syringe design shown in Figure 34, incorporates three soil sensors which insert into the soil collection chamber in the unexploded assembly. It also contains a syringe with 75 mL water that is passively plunged into the collection chamber via rotation of the payload bay which threads the ovoid-shaped object shown in Figure 34 around the central, non-rotating, and threaded lead screw. The syringe design utilizes the payload's rotation to dampen the soil to a moisture level of more than 30% so that the embedded sensors have the optimal conditions for testing. The addition of two extra sensors is a measure taken for redundancy and to take the averages of accurate values.

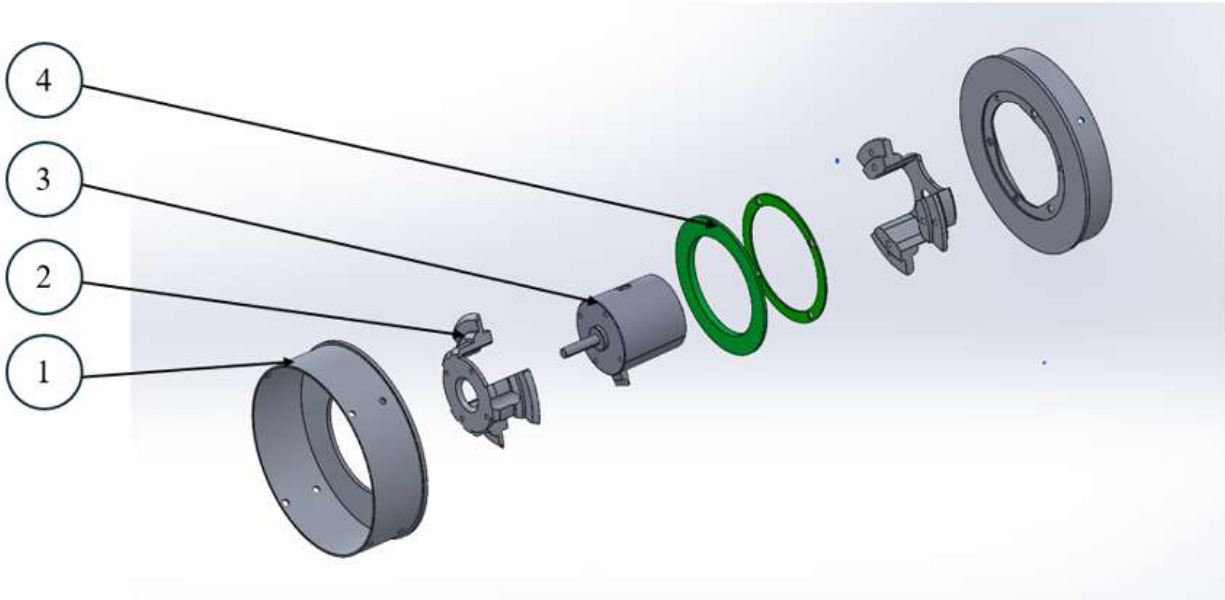


Figure 35: Rotating Bulk Plate System Exploded View. 1 represents a bulk plate in the rotating bulk plate system. 2 represents a collar connecting the motor to the bulk plates. 3 represents the motor used for all payload functions. 4 represents a Delrin disk that can act as a lighter substitute for a traditional bearing.

The Rotating Bulk plate System (RBS) can be viewed in Figure 35. This design was designed specifically to prioritize weight minimization because bulk plates can be quite heavy. The whole assembly weighs 2.88 lbs and will likely slightly decrease in weight when accounting for a smaller motor. This assembly is crucial for the rotating shovel design because it allows for the bulkplates in the payload to rotate freely while the rest of the aft section remains motionless.

4.3.5 Integrated Design

Landing Leg Deployment

At deployment of the drogue parachute, the landing legs, located fore in the vehicle of the soil collection and testing mechanism, will deploy passively. This is because the landing legs are spring-loaded and stowed within the airframe, with the tip of the leg resting on the intrados of the drogue bay airframe. At parachute deployment, they extend outward into their final position, which is shown on the right-hand side of Figure 36.

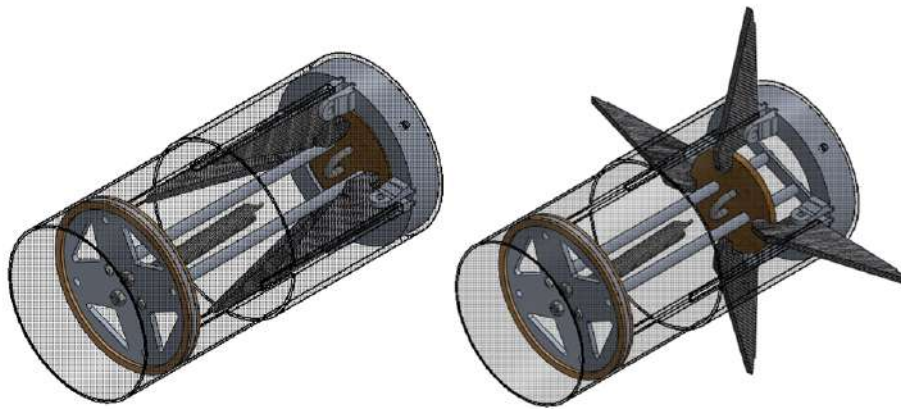


Figure 36: Landing Leg Geometry stowed (left) and deployed (right).

Payload Orientation

Upon landing, the vehicle will stabilize in a near-horizontal orientation, supported by the aft fins and landing legs as shown in Figure 37. Because the payload can rotate, the radial orientation of the rocket body about its longitudinal central axis is arbitrary and does not affect payload operations. The payload section, located between the aft fins and the landing legs, is structurally isolated using two bulkplates on either end, which will allow the section to rotate. The bulkplates also serve as the mechanical connection between the payload section and the rest of the vehicle.

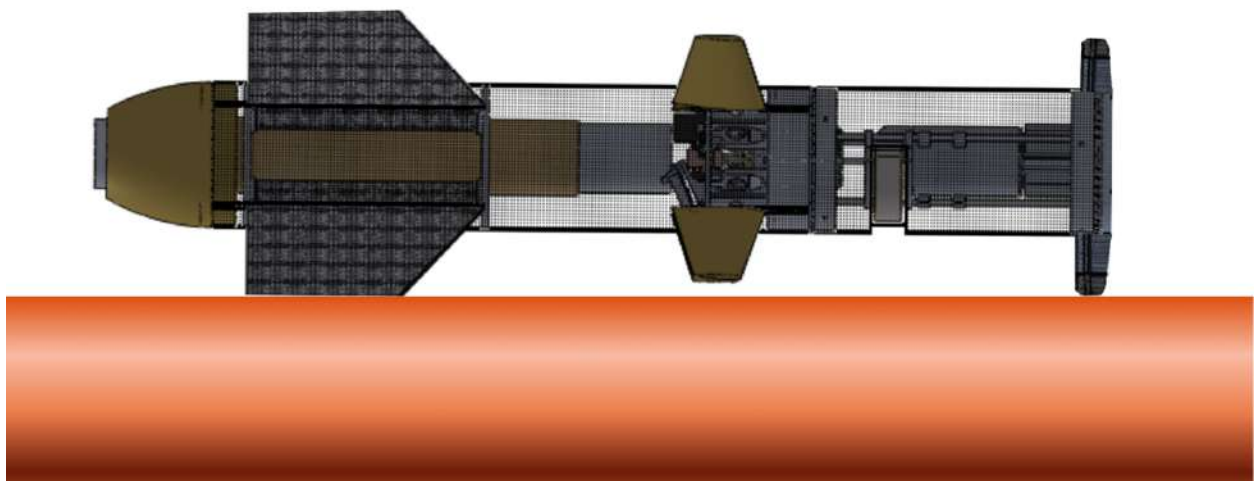


Figure 37: Stable landing configuration of the Vehicle.

Following landing, the drive motor, which is mounted aft of the payload section, will initiate through the Payload Electrical System, which begins a controlled rotation of the payload. The rotation drives shovel deployment, and after 48 rotations, soil contact will be achieved. To overcome soil resistance, the motor is rated at 23.8 Nm with a maximum angular velocity of 0.838 rad/s (8

rpm). The aft fins restrict the rotation of the vehicle to ensure that only the payload is rotating relative to the stationary lander structure, while the slightly pointed legs ensure that the whole aft section does not pivot about the fins due to the force of the soil on the shovel.

Shovel and Containment Chamber Subsystem

The shovel subsystem is designed to penetrate soil, collect it, and deliver it to the containment chamber for testing. As the shovel rotates and collects soil, gravity pulls the soil down a Teflon-coated slide into the cylindrical chamber fabricated out of Onyx composite, where it mixes with water and interfaces with soil sensors. The chamber has a usable internal volume of 310 mL.

Mechanically, the shovel system consists of the rotating bay, a stainless steel non-rotating lead screw, Kevlar wire, and a shovel body fabricated out of aluminum. The payload motor provides the only actuation for shovel deployment and collection bay rotation. During rotation, the Kevlar rope, which is wrapped around the non-rotating lead screw, translates rotational motion into linear extension to deploy the shovel outward to a maximum of about 3” external of the airframe, and nothing will protrude from the payload during flight per Nasa 4.6 requirement as the rotation will be locked in place via shear pins in the rotating bulkplate assembly which only break upon a very strong rotation force from the motor.

Water Delivery Subsystem

Simultaneously with soil collection, deionized water is injected into the containment chamber via an internal syringe system. The syringe plunger is linearly actuated by a carriage mounted on the non-rotating lead screw, leveraging the payload bay’s rotation for passive drive, as shown in Figure 34. Both the syringe body and plunger are machined from aluminum, selected for its low mass and ability to withstand hoop stresses generated under parachute shock loads well above 50 g. A check valve rated at 10 psi is mounted on the syringe to prevent water escape during the vehicle’s launch. The check valve rating was selected by calculating maximum water pressure, 3 psi, under parachute shock loads with a factor of safety of 3. The system is sealed using O-rings specified according to the Parker O-ring design guide, which provides an internal pressure rating of at least 20 psi. The syringe plunger discharges entirely after 48 rotations. This passive coupling eliminates the need for a dedicated actuator and ensures water delivery is synchronized with the soil acquisition process. The syringe has a capacity of 75 mL, sufficient to saturate a maximum sample of 150 mL of soil to 50% moisture content.

System Halt & Data Capture/Sensor Integration

After 48 rotations (over 8 minutes at 6 RPM), the drive motor halts for the soil testing stage. At this stage, the Payload Electrical System will initiate reading data from three soil sensors that are mounted inside the containment chamber. The three commercial off-the-shelf (COTS) seven-in-one soil sensors are statically mounted such that their prongs are slotted into the Onyx chamber and sealed using putty. Triplicated sensors provide redundancy and allow for data comparison to improve reliability.

Electrical System Overview

The Payload Electrical System's (PES) mission is to control the payload's action flow and record requisite data. The four primary goals of the payload are divided into subsystems being: The Payload Initiation Subsystem (PIS) which determines what phase of the mission the payload is in, the Payload Mechatronic System (PMS) which actuates the parachute detachment and payload drive motor, the Payload Sensor System (PSS) which records requisite soil sampling data, and the Payload Control System (PCS) which dictates the state-dependent system behavior and state transition criteria. The three software states are: Launch Runtime Standby, Soil Collection, and Soil Sensing. The PES logical state machine diagram is shown in Figure 38. The logical block diagram of the PES component connections and signals is shown in Figure 39.

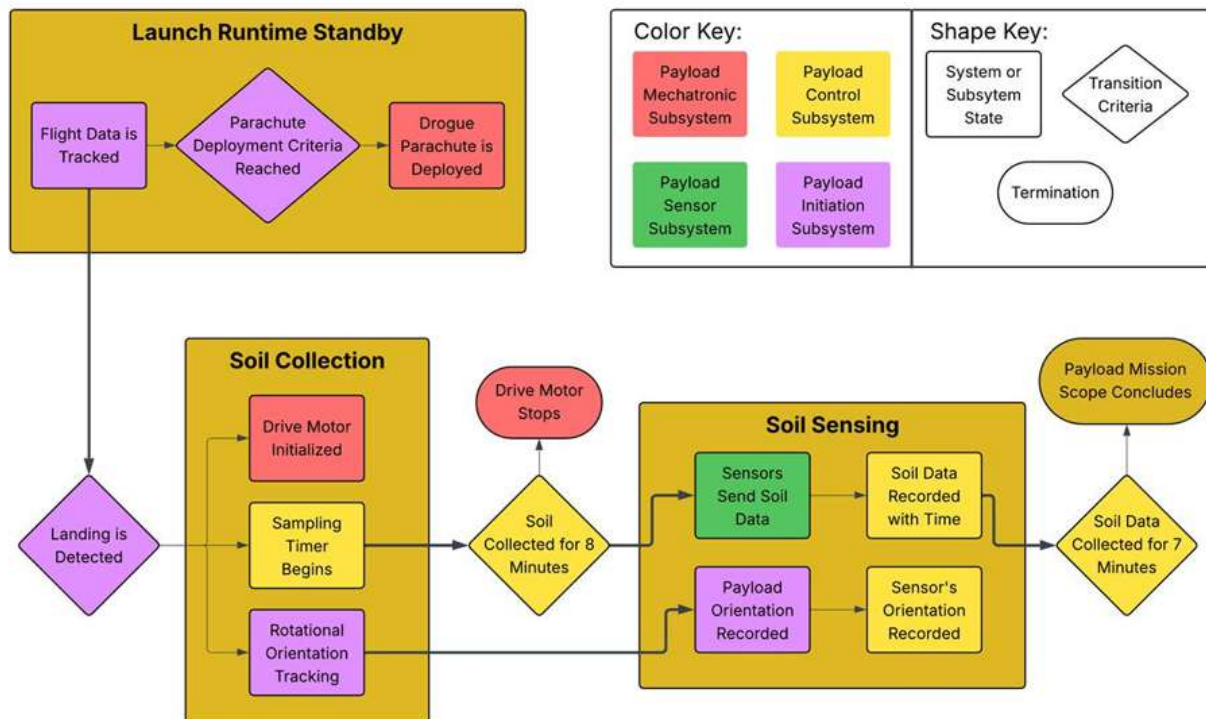


Figure 38: Payload Electrical System Logical State Machine Diagram.

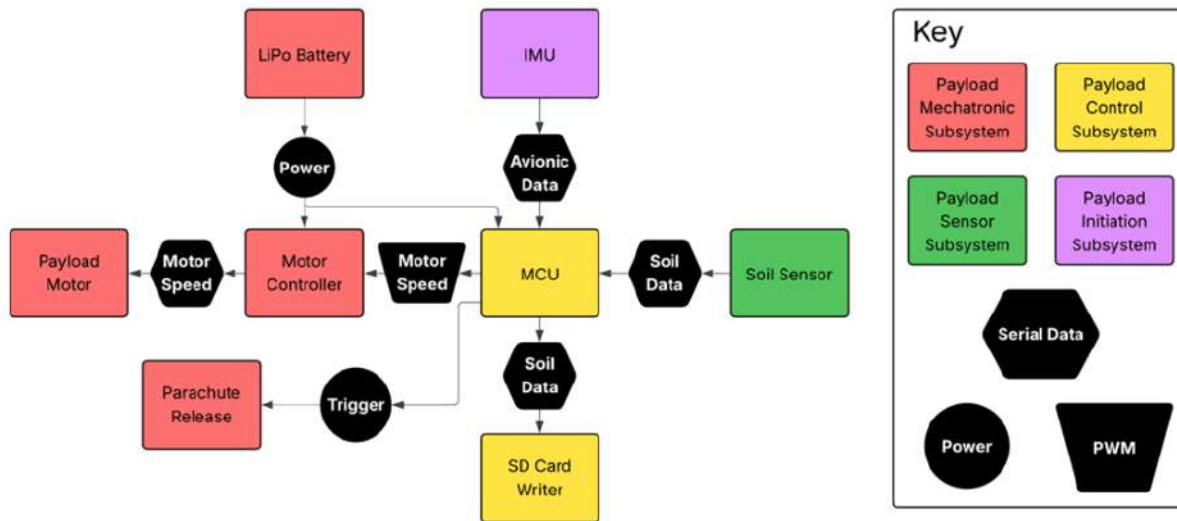


Figure 39: Payload Electrical System Logical Block Diagram.

Electrical System Component Choice

There are four primary electrical components for the PES which correspond to the four subsystems. This section will go over their functional methods and reasoning on their selection.

STM32F466RE Microcontroller

The PCS operates on the STM32F466RE microcontroller. It will run the central program that indicates when each component needs to perform what purpose. The STM32 was selected to be the main microcontroller for its low power consumption in relation to its high-speed embedded memory (512 kBytes) and number of peripheral interfaces (4 I2C and 6 UART).

RS485 Modbus Soil Sensor

The RS485 Modbus Soil Sensor is how the PSS will obtain soil characterization data. Timer-driven polling over Modbus RS-485 collects nitrogen, pH, and electrical conductivity levels at 200 Hz. Three sensors are incorporated in the collection chamber for different landing orientations. This sensor was chosen due to its simplicity, resiliency to the environment, and ease of integration with embedded systems.

VN-100 IMU

The PIS relies on the VN-100 IMU, which is configured to output flight telemetry (yaw, pitch, roll, acceleration (X, Y, Z), and pressure) at 200Hz, received via the STM32's UART interface. IMU readings are transferred using DMA to a double-buffer structure to ensure safe access. This flight data is used to detect specific flight conditions (launch, apogee, descent, and landing) and drives a finite state machine that directs the rest of the payload logic. This device was chosen for its proven reliability during testing, ease of integration with existing electrical hardware, and high-rate data output.

Brushed Motor and Planetary Gearbox

The 42GP-775 721:1 Reduction Planetary Gear and Brushed Motor is the component most crucial to the PMS operation. The motor is driven on 12V power by a PWM signal. We will revolve to 75% power at 6 rpm with 23.8 Nm of rated torque. This system overall was chosen to reduce weight as much as possible while simultaneously allowing for a theoretical torque limit that would enable flipping of the landed rocket to circumvent ground obstructions. Additionally, its built-in encoder enables us to determine the appropriate stopping point for shovel actuation after a set number of rotations, ensure the shovel both deploys and retracts properly.

STEMNaut Living Quarters

The STEMNaut Living Quarters serves as a crucial area of focus due to its importance for the survivability of the STEMNauts. This initial design is modeled after a Von Braun Wheel, which utilizes rotational motion during inter-planetary travel to create artificial gravity. In this way, STEMNaut could more effectively maintain their physical and psychological health during long term space travel. Additionally, STEMNauts can have direct access to the testing chamber, syringe, soil sensors and electronics, with access to the opening system, shovel, and motor close by outside of their atmospherically isolated compartment.



Figure 40: Preliminary Circular-Hollow STEMNaut Housing Unit.

4.4 Future Testing and Development Plans

Though the team has seen continued success with the initial payload prototype testing, there is still considerable work to be done before the payload, much less the rocket, is ready for flight. The current payload design relies on a Kevlar rope mechanism to open the shovel tip. Future development will include a gear-reducer coupled with a sector gear featured in Figure 33 discussed in Section 4.3.4. The intent for this design is that the high RPM motion of the motor will be slowed significantly so that the final pinion and sector gear rotate through the sector gears' outer

arc length after 48 rotations of the payload bay. Next steps in this design include rapidly producing a CAD model of this new extension design. Once the payload with the new opening mechanism is fabricated, testing will begin to validate the whole mechanism. This will include a load testing of the shovel, payload frame, and extension mechanism. Since it will be hard to validate design strength using traditional tensile stress-tests, the full-prototype will be tested on extremely hard soil (much harder than what is expected during competition launch day) to see if each mechanism holds up. Since the loading due to soil puncture will be much higher in this case, and since it will be distributed throughout the design in a similar way to how it would be during competition, it may inform potential failure points of the payload without using complicated simulations or load testing. Testing on other components like airframes, fins, bulkplates, and bearings will be done in the following way:

1. **Airframe** — Continue bolt tear-out, axial compression, and hoop stress on fiberglass fullscale airframe (6 inch).
2. **Fins** — Bending test with base of fin fixed, and load applied to tapered end. Load will increase until fin fails.
3. **Bulkplates** — Tensile test of avionics configuration (bulkplate with eyebolts bolted into airframe); load cell is attached to eyebolts and airframe is fixed. For special cases, like the Rotating Bulkplate System, tensile testing will also be performed mainly to see at which load the bolts clamping the assembly will fail.
4. **Bearings** — Delrin bearings will undergo compressive tests to ensure they will not yield during flight or motor actuation. Tests will utilize a universal testing machine, with a load cell. The thickness of the Delrin ring will be taken before and after a load 3 times higher than expected (due to bolt pre-load being applied to its front face) to see if yielding has occurred within a $3\times$ safety factor.

These preliminary tests will help the team understand where material can be taken off, and where it needs to be added. This will ensure that each subsystem and design is as efficient as possible, where new testing only serves to validate the success of the newly created design.

4.5 Validation Plan

To validate the shovel, soil collection chamber, shovel opening mechanism, and the payload in general, rigorous testing must be undertaken. The team needs to prove that the shovel can rotate continuously through at least 48 rotations with 24 passes and collect at least 50 mL of soil (Ideally closer to 90). A variety of soil conditions will need to be tested including nominal dry loose soil, but also muddy, hard, and rocky soils. Another facet of validation is ensuring that the payload can still successfully dig irrespective of the orientation in which it lands. The team predicts that with the axial symmetry of the rocket fins and legs coupled with the strong torque provided by the shovel, the payload should be able to reorient itself if the shovel rotates into rigid surface like a rock. That is to say, when the shovel digs into a rock, the shovel will act as a moment arm on the rocket body; if the force is adequate, it will cause the rocket body to tip about the stabilizing fins. The predicted torque required for flipping the payload onto another set of legs and fins is around 16.11 Nm (reference Section 4.3.3 for hand calculations). Failure testing on the threaded rods which station the payload connecting the shovel and soil collection chamber to the bulk plates will be important for understanding how the rods will behave under flight loads. An additional

key point of validation will be to ensure that the syringe included in the payload assembly does not release water into the soil collection chamber prematurely during flight. The payload will need to be properly cushioned to prevent too much play in the assembly connecting the soil sensors, syringe, and soil collection chamber because if any piece were to be dislodged, it would greatly hinder soil testing. A major point of validation is whether the motor can properly actuate multiple systems. Not only does it need to rotate the payload bay, but it will also provide torque for the syringe to plunge, and the shovel opening mechanism to open and close under stiff soil and rock loads. The team is confident the motor has ample torque due to its robust 23.8 Nm stall torque; however, further testing will validate the actual motor performance in the payload. The power requirement of the motor is a concern as the 14.8V LiPo batteries have considerable volume and mass. However, since the payload intends to be rotated so slowly (The LiPo batteries we are using have a charge capacity of 2200 mAh and weigh 226 grams). To test if the power draw will be significant, we will monitor the power used during a test of the payload in hard soil.

4.6 Secondary Payload Design

VADL has chosen to develop an experimental secondary payload that controls the drag of the rocket in flight to reach an accurate target apogee. The Apogee Control System (ACS) payload uses real-time altitude and velocity data to predict the apogee of the rocket, then deflects the mounted fins to induce drag. Fins deploy in pairs, opposing one another to only induce drag, instead of roll. While air brake systems have been used by USLI teams in the past, they primarily use drag plates or clam shell designs. These designs have been proven to be reliable but difficult to change. VADL's design allows more flexibility through fin iteration and replacement and a more streamlined design.

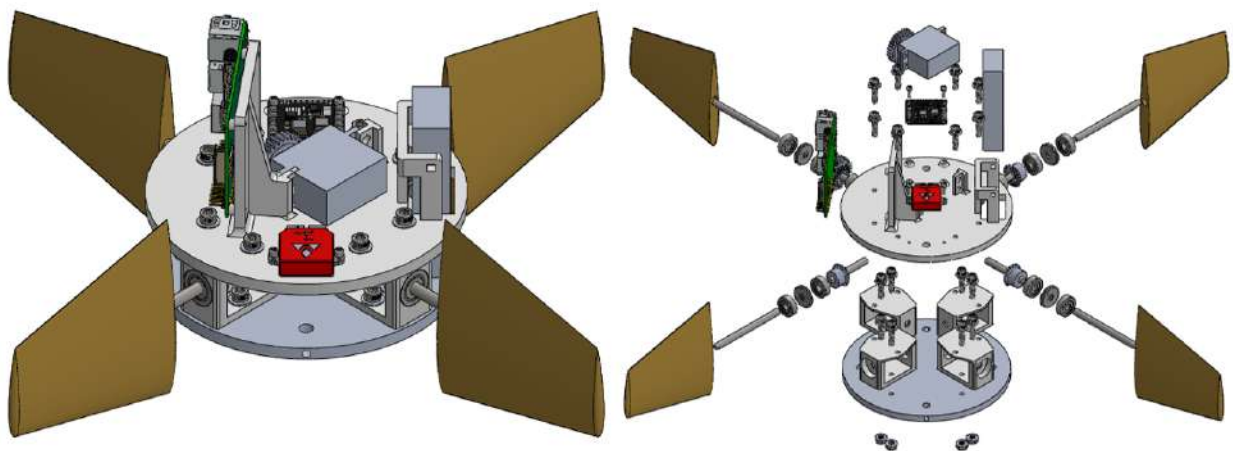


Figure 41: Collapsed (left) and exploded (right) view of preliminary ACS CAD. Features include VectorNav IMU, Raspberry Pi, Matek power hub, Lithium Polymer battery, 3D printed NACA 0012 airfoils, ball bearings, miter gears, spur gears, and a servo.

4.6.1 Integrated System Design

Attaining the target apogee is of high importance to the competition and thus the Apogee Control System is designed for reliability and simplicity. This payload consists of the fins, actuation system, and isolated electronics system. Mechanical and Electrical Subsystems are explained in detail

below.

Mechanical

The fins design is important because they are always deployed on the side of the rocket but are oriented parallel to the air during takeoff, minimizing the frontal area and drag of the rocket. The fins are NACA 0012 airfoils. During flight these fins rotate to increase the frontal area of the rocket, and thus the drag. To actuate the fins to only induce drag, the fins rotate in pairs towards each other. This motion is achieved with miter gears powered by a servo motor.

Electrical

A Raspberry Pi is used as the computer for this operation due to its commercial availability and reliability. Functioning as the system overhead, it initializes and powers the IMU and actuates the servo. The Raspberry Pi logs IMU data consisting of altitude, acceleration, and orientation. Additionally, it uses this data to continuously predict the apogee if it were to achieve the target apogee if the fins were deployed at that moment.

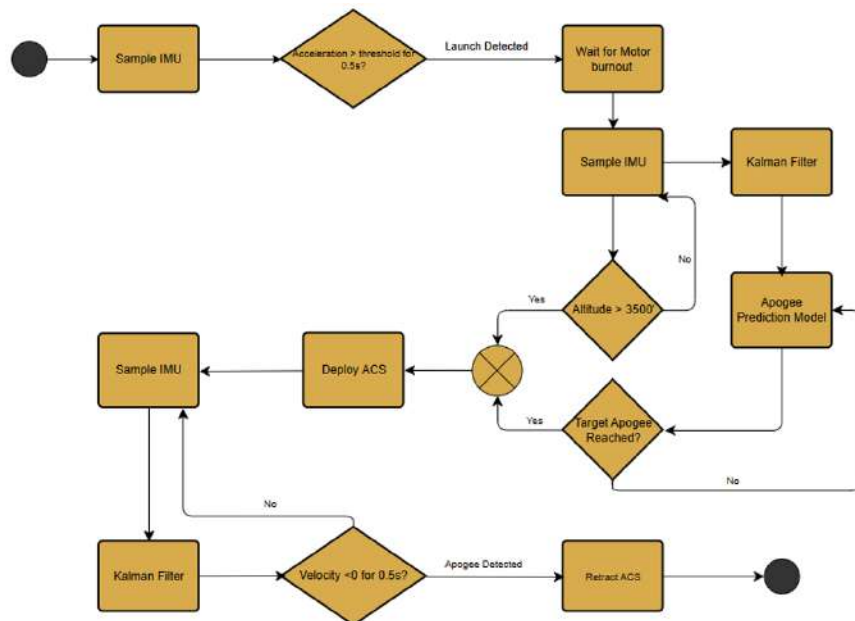


Figure 42: Apogee Control System flight operations.

4.6.2 Flight Dynamics

To understand the impact the Apogee Control System has on rocket flight, it was incorporated into VADL's MATLAB rocket simulation library: Declarative Rocket Simulation Software (DRSS). Detailed apogee prediction and trajectories under ACS deployment can be found under Section 4.2.4.

5 Educational Engagement Plan

5.1 Overview

At VADL, we will engage with the local community in order to promote our team's mission and inspire the next generation of students to pursue careers in STEM. We aim to deploy a wide variety of methods to achieve this goal. Our team will participate in STEM events throughout Nashville and surrounding areas to raise awareness and form industry connections; however, our main outreach priority will be inspiring and engaging with students in the local community.

We plan to do this by connecting with a wide variety of schools across many different neighborhoods to reach as many students as possible. One of our primary goals with this widespread outreach is to show students across all backgrounds that they can succeed as an engineer. We currently have a list of 70+ potential schools and have begun contacting them to plan events. We have already discussed with teachers and faculty about what type of lessons would be most beneficial to their students.

Some topics we plan to cover are as follows:

- What it means to be an engineer.
- Why the world needs engineers.
- How engineering affects everyday life.
- The NASA student launch project.
- Past, current, and future NASA projects and their importance.
- Engineering concepts like Newton's laws, electromagnetism, etc.
- Hands-on learning projects and competitions.
 - Building a rocket out of popsicle sticks.
 - Projectile motion tests and predictions.
 - Low-friction cart experiments.
 - Designing paper parachutes.

Detailed below is a condensed list of schools that we are either already in contact with or plan to connect with in the near future.

Elementary Schools

- Dan Mills Elementary School
- Julia Green Elementary School
- Harding Academy
- Whitsitt Elementary School
- Crieve Hall Elementary School

Middle Schools

- Two Rivers Middle School
- J.T. Morre Middle School
- West End Middle School

High Schools

- McGavock High School
- Franklin High School
- White's Creek High School

K-12 Schools

- Merrol Hyde Magnet School
- Nashville Christian School
- Franklin Road Academy

5.2 Lesson Plan Details

Below is a sample lesson plan:

1. Introduce our team.
2. Simple quiz at the beginning (basic questions about prior knowledge of STEM).
3. What is Engineering?
 - Ask students what they know about engineering.
 - Discuss what comes to mind when they hear “engineering”.
 - Teach them about different engineering disciplines.
 - Explain why engineering is important.
4. Explain our project and the rocket.
 - Describe project scope and goals.
 - Explain stages of the launch.
 - Discuss different sections of the rocket and their functions.
 - Show pictures and videos of design/testing process.
5. Teach them about NASA.
 - NASA's mission statement and role in society.
 - Past, present, and future NASA projects.
 - Future need for engineers at NASA.
6. Explain activity (give directions, distribute supplies, form teams).
7. Do activity (collaborative hands-on projects).
8. Wrap-up discussion (connect activity to engineering concepts, emphasize future opportunities).
9. Quiz at the end (evaluate learning outcomes, prompt brainstorming on next steps in STEM).

5.3 Outreach Benefits

Earlier, we mentioned that our goal is to impact and inspire the next generation of STEM students. VADL is deeply invested in this mission, and we believe that this is one of the cornerstones of this project. Often, students don't have the opportunity to witness STEM in practice. They may take a generic science course or learn how to perform basic arithmetic, but it can be difficult for them to see the real-world applications. We want to illustrate how STEM can be applied in their lives by walking them through the design of our rocket and connecting it to basic concepts that they have already learned. Additionally, we will provide general examples of how engineering shows up in their everyday lives. By the end of the lesson, they will recognize that almost everything manmade exists because of the work engineers have done.

Another main objective is to prove to students that anyone is capable of becoming an engineer. We want to emphasize that someone from any background, culture, or gender can have success in the STEM world. We will do this by telling inspirational stories from engineers who come from diverse backgrounds. Additionally, we will display the different types of people within our own team. We hope every student we reach feels represented within our team.

The final goal of our educational plan is to convince students that STEM can be fun. STEM often gets a bad reputation for being boring, difficult, or inaccessible to the layperson. We want to disprove this bad reputation by showing students that not only is it fun, but it is also interesting. To accomplish this, we will explain why our members love STEM and have them participate in hands-on activities. The aim is to spread the passion of our members to the students. The activities we've selected were specifically curated so students can easily engage and enjoy themselves. Some examples are launching stomp rockets to demonstrate aerodynamics, designing a rocket from popsicle sticks to emphasize the different components of a rocket, and performing experiments with low-friction carts to explain some of Newton's laws. It is vital to show the students that they can engage in STEM no matter where they are in their educational journey. By giving them a small taste of these types of experiments, we will encourage them to seek more STEM activities and opportunities. At the end of our presentation, we will give the students actionable examples of how they can expand their STEM knowledge further. For example, watching fun videos made by STEM creators (Mark Rober, Bill Nye, etc.), creating things with paper and cardboard, playing with toys like snap circuits, going to the museum, or joining a science-related club.

In conclusion, community engagement is a vital part of our mission because we can impact the future of hundreds of students. Every single person on our team has had at least one moment in their lives that inspired them to become engineers. With these presentations, we strive to create this moment for every student we reach. We are not only looking to influence their career goals, but also their lives. Becoming an engineer is a life-altering experience due to the impact your projects can have on the world, and we want to empower students to pursue this opportunity.

6 Project Plan

6.1 Timeline

Success in the USLI competition requires a detailed schedule that covers meetings, report writing, builds, testing, travel, and all review gates. VADL manages day-to-day work in Jira with an Agile cadence while tracking the project at the milestone level in a waterfall layout. External commitments (Proposal, PDR, CDR, FRR/FRR-A, LRR/Launch, PLAR) are modeled as fixed-date milestones; internal delivery runs in two-week sprints with Epics and tasks that roll up to those milestones. More details regarding project management at the work level can be found in Section 1.9.

The milestone Gantt anchors the project around hard due dates and windows for Proposal, PDR, CDR, FRR/FRR-A, subscale/VDF/PDF, Launch Week (or home-field launch window), and PLAR. Each milestone block includes its submission deadline, NASA Q&A/VTC window, and any planned travel or field operations.

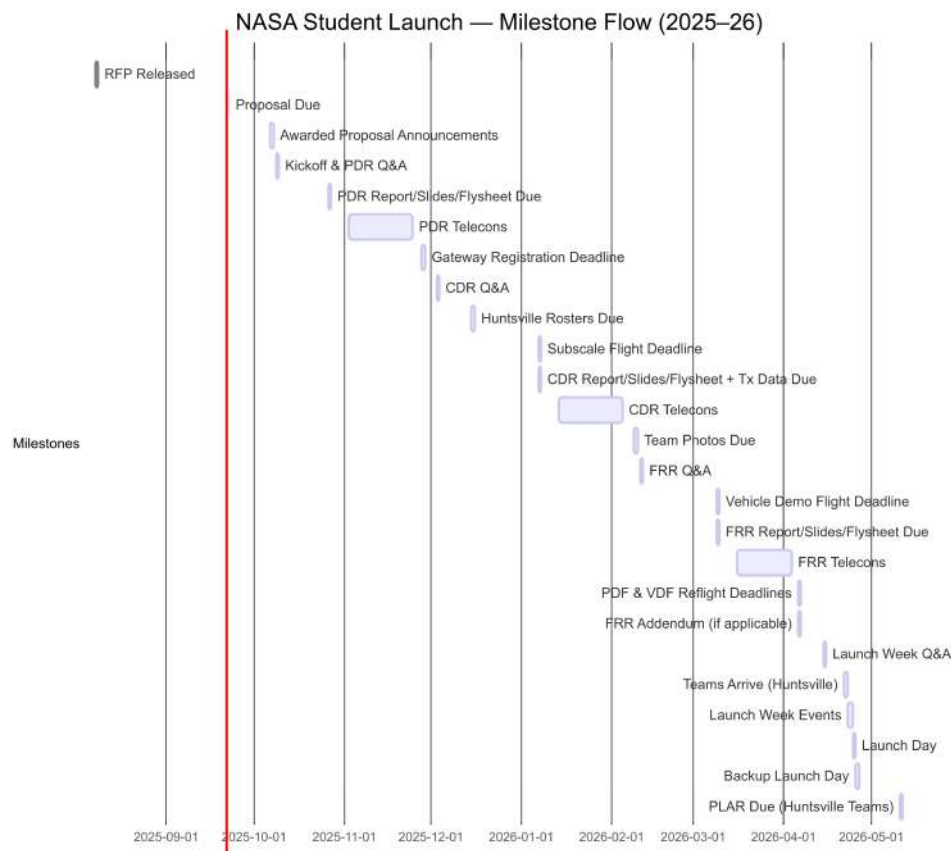


Figure 43: VADL 2025-2026 Milestone Gantt Char.

6.2 Detailed Budget

Our projected budget for the 2025-2026 NASA student launch team is \$15,000. We came up with this number based on our predicted costs based on previous projects, calculated projections,

and our estimated fundraising. In order to ensure our money is spent wisely, we track all our purchases in Excel and require approval before purchases are made. We are also intentional in taking advantage of materials that we can access from past projects or from other labs at Vanderbilt. We recognize the uncertainty of a long-term project such as this one, so we calculated what we believe to be a necessary emergency budget in case any unforeseen problems arise. We made cost predictions of the payload and vehicle based on the parts we expect to need and based on expenses from past projects. Finally, we created a detailed plan regarding the budget for our trip to the competition, as well as any other trips needed. See the figure below for a detailed breakdown of our budget.

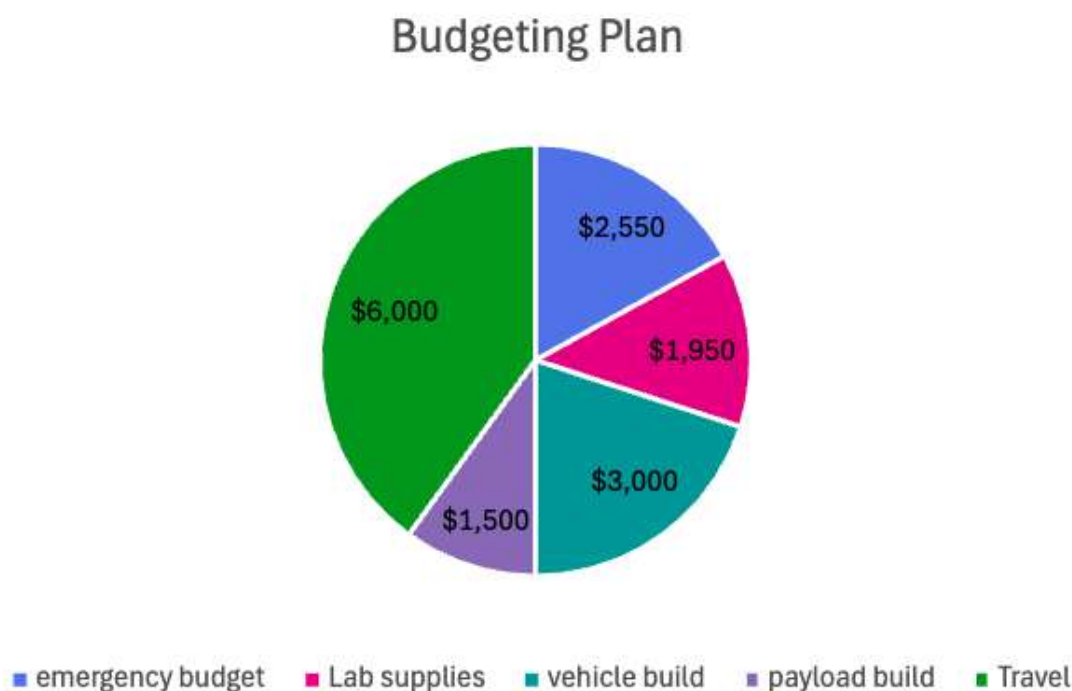


Figure 44: Proposed budget plan pie chart.

6.3 Fundraising Plan

We are still finalizing our funding plan; however, we were able to make a strong prediction based on the fundraising our team has done in previous years. So far, we have already been granted \$5000 from the Vanderbilt Engineering Dean's Fund. We plan to source the rest of our funding from Vanderbilt alumni, the Dalton VADL Fund, and leftover funding from past years. See the figure below for a detailed breakdown of our budget.

Fundraising Plan

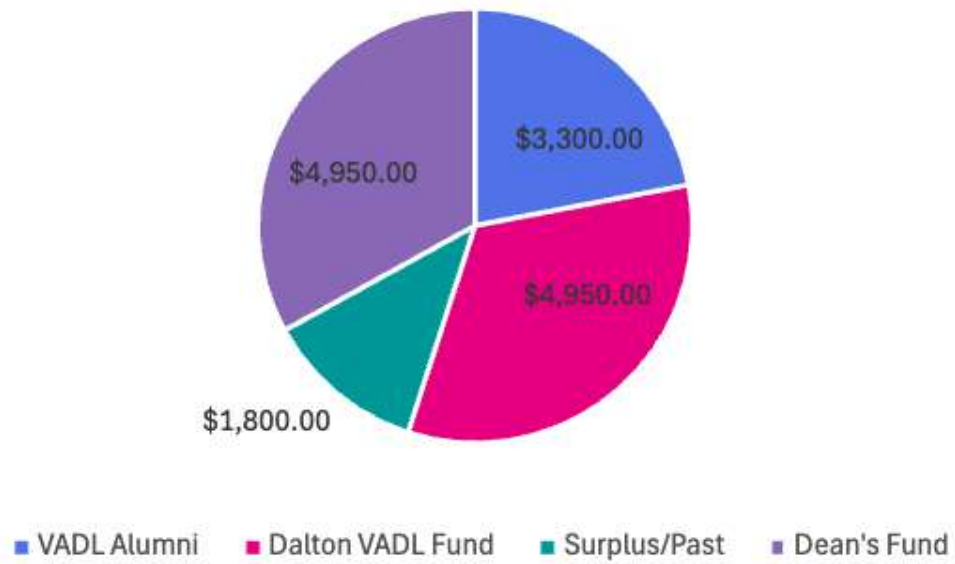


Figure 45: Proposed budget plan pie chart.

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